Report on

Database for Smart City Information Management System

April 10, 2026

Prepared By:

2211275042, Hasnat Karibul Islam
2211022042, Md. Minhajul Islam
2212021042, Nur Ibna Kawsar Zitu
2212642042, Md. Sabbir Hossain

Course Instructor:

Prof. Dr. Kamruddin Nur
Professor, Dept. of Electrical and Computer Engineering

Contents

1 Project Title **9**

2 Project Description **9**

3 Project Objective **10**

4 Deliverable **11**

5 Diagrams **12**

5.1 Project Schedule . . . 12
5.2 Conceptual Design Diagram . . . 13
5.3 Logical Design Diagram . . . 14
5.4 Physical Design Diagram . . . 15

6 Table Creation **16**

7 Queries **31**

7.1 Trigger . . . 31
7.2 Adding the Foreign Key Constraint . . . 31
7.3 Citizen Registration Approval . . . 31
7.4 Citizen Registration Rejection . . . 32
7.5 Registration Applicants with Pending Status . . . 32
7.6 Citizens Registered This Month . . . 32
7.7 All Department Name and ID . . . 34
7.8 Top Performing Departments in IoT Usage . . . 35
7.9 Service Provided by Each Department . . . 35
7.10 Department Head . . . 36
7.11 List of Services with Feedbacks . . . 37
7.12 Counting Feedbacks by Services . . . 38
7.13 List Traffic Data by IoT Device . . . 39
7.14 Location with High Vehicle Count . . . 40
7.15 Locations where No Citizens Live . . . 41
7.16 Counting Citizens in Each Area . . . 42
7.17 List of Citizens with Their Location and City . . . 43
7.18 List Maintenance Activities by Department . . . 43
7.19 Counting Maintenance Activities per Department . . . 44
7.20 List of Ongoing Maintenance in Each Location . . . 44
7.21 Location with Low Vehicle Count . . . 45
7.22 Finding All Requests by Citizens Along With Their Names . . . 45
7.23 Finding Officials with the Most Pending Requests . . . 46
7.24 Finding Locations with Highest Number of Requests . . . 46
7.25 Retrieving Unresolved Feedback . . . 47
7.26 List Available Services . . . 48
7.27 Services with a Fee Above 500 Taka . . . 48
7.28 Services with a Fee Below 500 Taka . . . 49
7.29 Counting Service Requests by Status . . . 49

7.30 Finding Locations with the Most Frequent Services Requested and Assigned Official Details . . 50
7.31 List of Locations Experiencing Extreme Weather 50
7.32 Average Temperature and Humidity by Weather Condition 51
7.33 Monthly Average Weather Statistics 51
7.34 City Official Details who Don't Have any Supervisors 52
7.35 Most Frequent Routes, Average Number of Stations and Status Distribution 52
7.36 Bus Route and Frequency Analysis 53

8 Conclusions **53**

9 Acknowledgments **54**

List of Figures

1 Gant Chart . . . 12
2 Conceptual design diagram of the database . . . 13
3 Logical design diagram of the database . . . 14
4 Physical design of the database . . . 15
5 Citizen Registration Table . . . 16
6 Citizen Info Table . . . 17
7 City Official Table . . . 18
8 Department Table . . . 19
9 Public Service Table . . . 20
10 Feedback Table . . . 21
11 Service Request Table . . . 22
12 Location Table . . . 23
13 Citizen Location Table . . . 24
14 Weather Condition Table . . . 25
15 Infrastructure Maintenance Table . . . 26
16 Bus Transportation Table . . . 27
17 IoT Devices Table . . . 28
18 Traffic Data Table . . . 29
19 Utility Monitoring Table . . . 30
20 Pending status applicant Table . . . 32
21 Citizens Registered This Month Table . . . 32
22 Department Name and ID Table . . . 34
23 Top performing departments in IoT usage . . . 35
24 Service and Department Table . . . 35
25 Department Head Table . . . 36
26 Services with Feedbacks Table . . . 37
27 Counting Feedbacks by Services Table . . . 38
28 Traffic Data by IoT Device Table . . . 39
29 Location and high vehicle count table . . . 40
30 Locations where no citizens live Table . . . 41
31 Counting citizens in each area and location table . . . 42
32 Citizens location and city table . . . 43
33 Maintenance Activities by Department table . . . 43
34 Maintenance Activities per Department count table . . . 44
35 Ongoing maintenance in each location table . . . 44
36 Location with low vehicle count table . . . 45
37 All Requests by Citizens Along With Their Names Table . . . 45
38 View all the Officials with the Most Pending Requests . . . 46
39 Locations with highest number of request Table . . . 46
40 Unresolved Feedback Table . . . 47
41 Available Services Table . . . 48
42 Services with a Fee Above 500 Taka Table . . . 48
43 Services with a Fee Below 500 Taka Table . . . 49

44 Counting Service Request by Status Table 49
45 Locations with the Most Frequent Services Requested and Assigned Official Details Table . . . 50
46 Locations Experiencing Extreme Weather Table 50
47 Average Temperature and Humidity by Weather Condition Table 51
48 Monthly Average Weather Statistical Table 51
49 City Officials details who don't have any supervisors Table 52
50 Most Frequent Routes Average Number of Stations and Status Distribution Table 52
51 Bus route and frequency analysis Table 53

List of Tables

1 Team Member's Contributions 8

Project Overview

The Smart City Information Management System (SCIMS) project is an advanced, integrated platform designed to manage and optimize urban infrastructure and public services using modern technologies like the Internet of Things (IoT), AI, big data analytics, and cloud computing. This system plays a crucial role in addressing the increasing demands on cities due to rapid urbanisation, helping cities become more efficient, sustainable, and resilient. By deploying IoT sensors across a city's infrastructure—such as transportation networks, energy grids, water supply systems, and waste management facilities—SCIMS continuously collects real-time data. This data is then processed using AI and advanced analytics, allowing city officials to make informed, data-driven decisions that optimise city operations. For instance, predictive algorithms can manage traffic flow based on real-time conditions, reduce energy consumption by balancing supply and demand, and optimise waste collection routes based on actual container fill levels. A core component of SCIMS is a centralised management platform providing city authorities with a holistic view of all urban infrastructure. This enables seamless coordination between various departments, improving operational efficiency and reducing silos that traditionally hinder effective city management. SCIMS also supports predictive maintenance, where infrastructure issues such as wear and tear on roads, bridges, or water systems are identified and addressed before they escalate into more significant, costlier problems. Additionally, the system is designed with sustainability in mind, optimizing resource usage and promoting environmentally friendly practices. For example, smart grids efficiently manage energy consumption, while smart waste management systems help reduce carbon emissions by optimizing collection schedules and routes. The benefits of SCIMS are wide-ranging. It significantly enhances the quality of life for residents by improving public services like transportation, water distribution, and energy management, making cities more livable. It also offers significant cost savings through operational efficiencies and preventive maintenance, reducing the need for expensive, reactive repairs. Moreover, SCIMS contributes to environmental sustainability by minimizing energy waste, lowering emissions, and improving resource efficiency. Overall, SCIMS helps cities operate more smoothly in the present and positions them for future growth, creating smart, sustainable urban environments capable of withstanding the challenges of the 21st century.

Contributions

Table 1: Team Member's Contributions

ID	Name	Tasks	Contribution
2211275042	Hasnat Karibul Islam	• Project Planning • Project Overview • Project description • Project Objective • Conceptual and Physical Diagram • Gantt Chart • Logical Diagram • Table Creation • Query design and Implementation • Project Debug and testing	33%
2211022042	Md. Minhajul Islam	• Project Planning • Project Overview • Project Deliverable • Logical Diagram • Data Populating • Query • Project Debug and testing	32%
2212021042	Nur Ibne Kawsar Zitu		25%
2212642042	Md. Sabbir Hossain		10%

1 Project Title

Database for the Smart City Information Management System

2 Project Description

The Smart City Information Management System (SCIMS) is an advanced database-driven solution designed for streamline and to enhance the management of modern urban operations. We have developed this system using MySQL. **SCIMS** serves as a centralized platform for integrating, monitoring and optimizing various aspects of city administration, ensuring data transparency, resource efficiency, and improved decision-making processes. **SCIMS** also support us in critical city functions through its comprehensive database structure. The **CitizenRegistration** and **CitizenInfo** tables manage citizen profiles and automate workflows for registrations, ensuring compliance with policies such as age-based eligibility. The **PublicServices**, **ServiceRequest**, and **Feedback** tables streamline the delivery of city services by tracking service requests, feedback, and resolutions, fostering accountability and citizen satisfaction. Transportation management is enhanced using the **BusTransportation** table, which tracks routes, operational status, and service frequency. Location-based data, stored in the **Location** table and linked to citizen demographics and infrastructure, provides actionable insights for planning and resource allocation. The **InfrastructureMaintenance** and **UtilityMonitoring** tables log utility consumption, maintenance schedules, and costs, promoting efficient resource utilization and sustainability. **SCIMS** also integrates environmental monitoring with the **WeatherConditionData** table, providing weather insights crucial for emergency preparedness and urban planning. IoT integration is facilitated through the **IoTDevices** and **TrafficData** tables, enabling real-time monitoring of traffic, energy consumption, and other urban metrics. By centralizing all the data and fostering interconnectivity between city departments, **SCIMS** empowers administrators to identify trends, predict challenges, and implement proactive solutions. It also enhances citizen engagement by efficiently managing feedback and requests. Scalable and adaptable, **SCIMS** is designed to evolve with the city's growing needs, supporting future integration of emerging technologies. Overall, **SCIMS** is a transformative tool that drives smart governance, sustainability, and an improved quality of life for citizens.

3 Project Objective

The objectives of this project are:

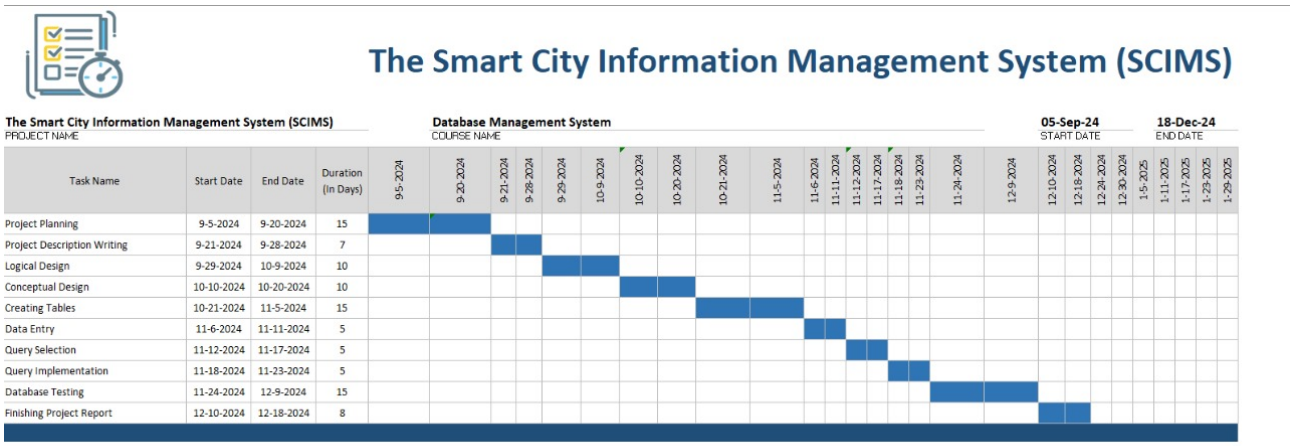
- **Centralized Data Integration:** Creating a comprehensive MySQL based database that will act as a centralized repository for controlling act of urban operation, enabling seamless data integration and access.
- **Citizen-Centric Services:** Streamline citizen interaction by efficiently handling profiles, registration, and service request using the **CitizenRegistration** and **CitizenInfo** tables, encouraging ease and inclusion.
- **Data-Driven Decision Making:** By offering real-time information regarding transportation, infrastructure, public services, and utility usage, urban managers may make better informed decisions, resulting in more competent government.
- **Operational Transparency and Accountability:** Foster trust by ensuring transparent city operations through well-organized tables like **PublicServices**, **Feedback**, and **ServiceRequest**, offering clear insights into resource utilization and service delivery.
- **Smart Urban Planning:** Utilize data from the Location, **WeatherConditionData**, and **Infrastructure-Maintenance** tables to support comprehensive urban planning and development initiatives.
- **Regulatory Compliance:** Automate governance policies, such as age-based eligibility for citizen registrations, with workflows and triggers that enforce compliance and streamline administrative processes.

4 Deliverable

- **A comprehensive and normalized MySQL database schema, encompassing the following tables: CitizenRegistration** and **CitizenInfo** for citizen management. **CityOfficial** and **Department** for administrative roles and departmental organization. **PublicServices**, **Feedback**, and **ServiceRequest** for service tracking and citizen feedback management. **Location**, **CitizenLocation**, **WeatherConditionData**, and **InfrastructureMaintenance** for location-based data and urban planning. **BusTransportation**, **IoTDevices**, **TrafficData**, and **UtilityMonitoring** for real-time monitoring and management of transportation, utilities, and smart technologies.
- **Automated Workflows and Triggers:** Implementation of triggers to automate workflows such as age-based approval of citizen registrations, efficient data updates, and compliance checks.
- **Citizen-Centric Features:** Modules for citizen registration, profile updates, and seamless interaction with city services, supported by structured and interconnected tables.
- **Comprehensive Reporting and Analytics:** Advanced SQL queries for generating actionable insights, such as service usage statistics, resource allocation trends, and system performance metrics.
- **Real-Time Monitoring Capabilities:** Tables and queries designed to monitor traffic **TrafficData**, utilities **UtilityMonitoring**, and IoT devices **IoTDevices** for maintaining optimal city operations.

5 Diagrams

5.1 Project Schedule



Task Name	Start Date	End Date	Duration (In Days)
Project Planning	9-5-2024	9-20-2024	15
Project Description Writing	9-21-2024	9-28-2024	7
Logical Design	9-29-2024	10-9-2024	10
Conceptual Design	10-10-2024	10-20-2024	10
Creating Tables	10-21-2024	11-5-2024	15
Data Entry	11-6-2024	11-11-2024	5
Query Selection	11-12-2024	11-17-2024	5
Query Implementation	11-18-2024	11-23-2024	5
Database Testing	11-24-2024	12-9-2024	15
Finishing Project Report	12-10-2024	12-18-2024	8

Figure 1: Gant Chart

5.2 Conceptual Design Diagram

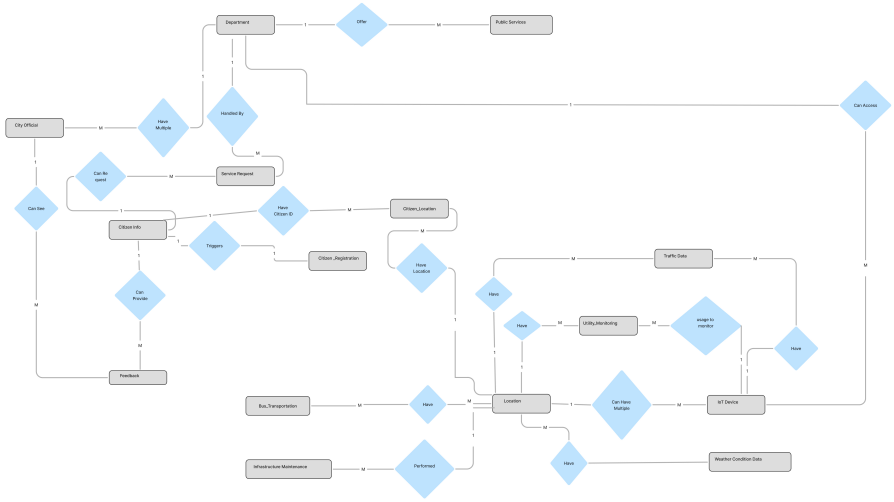


Figure 2: Conceptual design diagram of the database

5.3 Logical Design Diagram

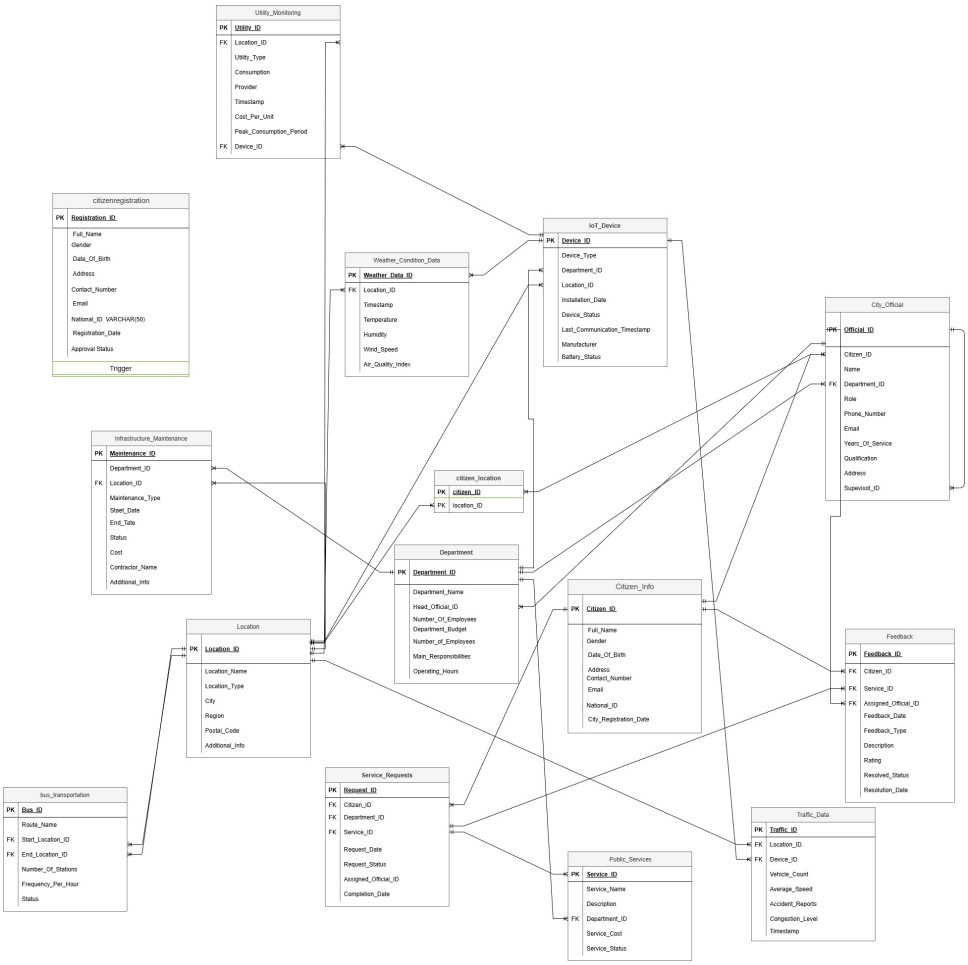


Figure 3: Logical design diagram of the database

5.4 Physical Design Diagram

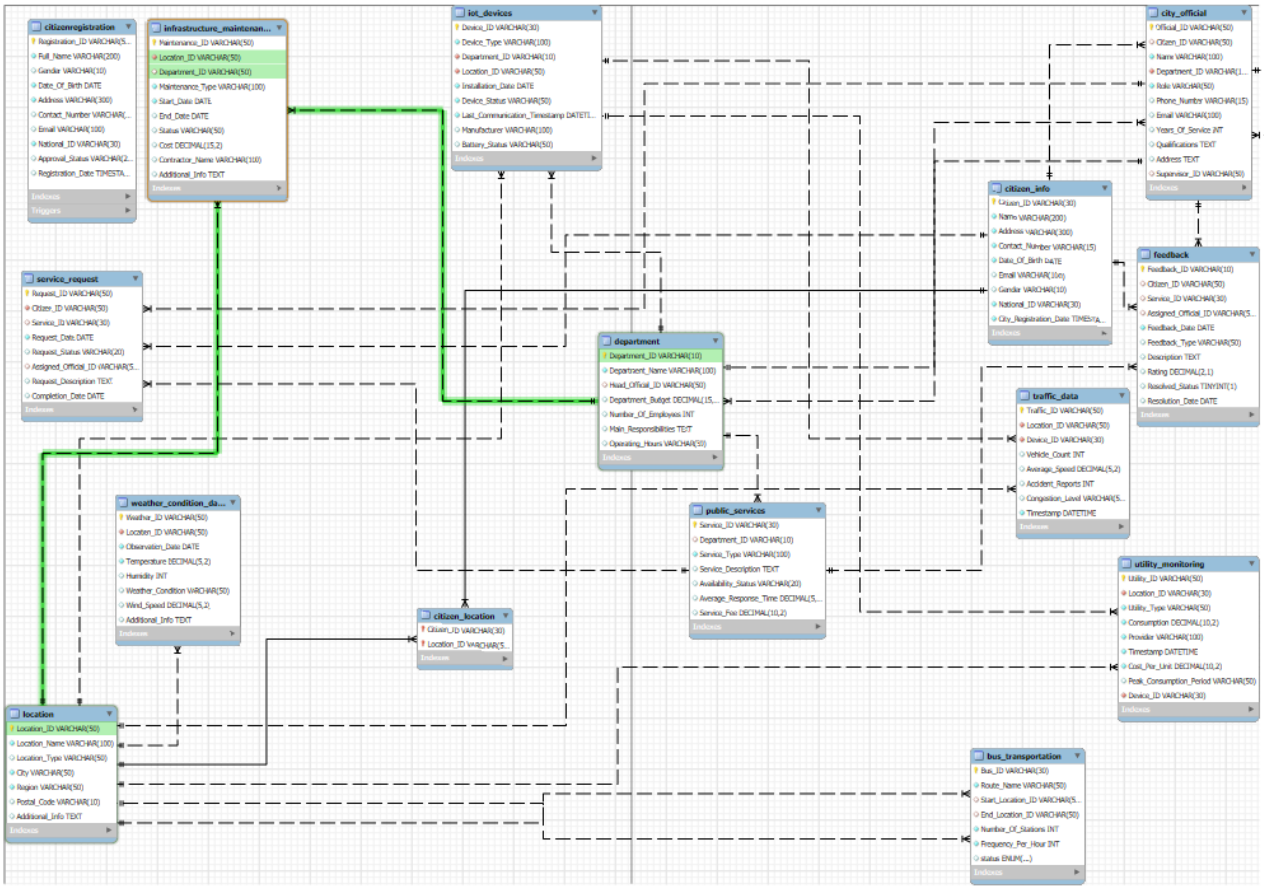


Figure 4: Physical design of the database

6 Table Creation

- **Citizen Registration:**

Registration_ID	Full_Name	Gender	Date_Of_Birth	Address	Contact_Number	Email	National_ID	Approval_Status	Registration_Date
CIT001	Professor Dr. Kamruddin Nur	Male	1998-10-18	House: 91; Road:04; Block:B, Banani Dhaka-1213	01777354323	kamruddin.nur@gmail.com	1997054672345	Approved	2024-12-08 05:12:31
CIT002	Hasnat Karibul Islam Alindo	Male	2001-12-28	House 526, Road 17, Block J, Bashundhara R/A	01316619278	withalindo@gmail.com	1998034530678	Approved	2024-12-08 05:12:31
CIT003	Md. Minhaujul Islam	Male	2001-04-16	House 257, Road 5, Block C, Bashundhara R/A	01756619289	minhajulislamrimon28@gmail.com	1990027530526	Approved	2024-12-08 05:12:31
CIT004	Nur Ibne Kawsar Zitu	Male	2000-11-15	House 226, Road 5, Block C, Bashundhara R/A	01712345670	nur.zitu@gmail.com	1990045530425	Approved	2024-12-08 05:12:31
CIT005	Sabbir Hossain	Male	2000-04-09	House 126, Road 15, Block A, Bashundhara R/A	01798766832	sabbirsts420@gmail.com	1990025530223	Approved	2024-12-08 05:12:31
CIT006	Sayed Mahi Ashrafi	Male	2001-09-17	House 26, Road 2, Block D, Nikunja R/A	01798765432	syed.mahi@gmail.com	1990067530124	Approved	2024-12-08 05:12:31
CIT007	Ivy Khan	Female	2001-08-06	House 427, Road 8, Block D, Bashundhara R/A	01745678901	ivy.khan@gmail.com	1990024498128	Approved	2024-12-08 05:12:31
CIT008	Meher Easha	Female	2001-08-15	House 126, Road 8, Block C, Bashundhara R/A	01711223344	emeherun@gmail.com	1990024530129	Approved	2024-12-08 05:12:31
CIT009	Md. Tanvir Hossain	Male	1999-10-26	House 15, Road 10, Gulshan 1, Dhaka	01712345678	tanvir.hossain@gmail.com	1990024530123	Approved	2024-12-08 05:12:31
CIT010	Nasima Akhter	Female	1999-05-18	Flat 5C, Lakeview Apartments, Banani, Dhaka	01812345679	nasima.akhter@yahoo.com	1990054525678	Approved	2024-12-08 05:12:31
CIT011	Shahriar Kabir	Male	1990-08-26	House 24, Road 8, Dhanmondi, Dhaka	01912345670	shahriar.kabir@gmail.com	1998014530987	Approved	2024-12-08 05:12:31
CIT012	Ayesha Rahman	Female	1996-01-26	House 32, Sector 7, Uttara, Dhaka	01798765432	ayesha.rahman@gmail.com	1995056783456	Approved	2024-12-08 05:12:31
CIT013	Md. Abir Hasan	Male	1987-10-06	House 18, Road 5, Mohammadpur, Dhaka	01876543210	abir.hasan@outlook.com	1996034526789	Approved	2024-12-08 05:12:31
CIT014	Taslima Begum	Female	1975-10-26	Flat 4A, Green Plaza, Mirpur 12, Dhaka	01745678901	taslima.begum@gmail.com	1993034786541	Approved	2024-12-08 05:12:31
CIT015	Khaled Mahmud	Male	1985-12-16	House 22, Bashundhara R/A, Block F, Dhaka	01899887766	khaled.mahmud@gmail.com	1998012567894	Approved	2024-12-08 05:12:31
CIT016	Shivan Sharma	Male	1985-12-16	Flat No. 102, Block A, Greenwood Apartments, ...	01559887766	shivam.sharma@gmail.com	1988012567894	Rejected	2024-12-08 05:12:31
CIT017	Raiyan Ahmed	Male	2010-12-16	House 22, Banasree, Block C, Dhaka	014589887766	raiyan.ahmed@gmail.com	1998012567895	Approved	2024-12-08 05:12:31
CIT108	Reshman Nuriva	Female	2002-01-06	House 158 Dhanmondi-7, Dhaka	01405872035	reshman.nuriva@outlook.com	1998012567492	Approved	2024-12-10 04:57:54
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 5: Citizen Registration Table

```sql
CREATE TABLE CitizenRegistration (
    Registration_ID VARCHAR(500) PRIMARY KEY,
    Full_Name VARCHAR(200) NOT NULL,
    Gender VARCHAR(10) CHECK (Gender IN ('Male', 'Female', 'Others')),
    Date_Of_Birth DATE NOT NULL,
    Address VARCHAR(300) NOT NULL,
    Contact_Number VARCHAR(15),
    Email VARCHAR(100) UNIQUE,
    National_ID VARCHAR(30) UNIQUE NOT NULL,
    Approval_Status VARCHAR(20) CHECK (Approval_Status IN ('Pending', 'Approved'
    , 'Rejected')) DEFAULT 'Pending',
    Registration_Date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

- **Citizen Info:**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Citizen_ID	Name	Address	Contact_Number	Date_Of_Birth	Email	Gender	National_ID	City_Registration_Date
CIT001	Professor Dr. Kamruddin Nur	House: 91; Road:04; Block:B, Banani Dhaka-1213	01777354323	1998-10-18	kamruddin.nur@gmail.com	Male	1997054672345	2024-12-08 05:12:31
CIT002	Hasnat Karibul Islam Alindo	House 526, Road 17, Block J, Bashundhara R/A	01316619278	2001-12-28	withalindo@gmail.com	Male	1998034530678	2024-12-08 05:12:31
CIT003	Md. Minhaujul Islam	House 257, Road 5, Block C, Bashundhara R/A	01756619289	2001-04-16	minhajulislamrimon28@gmail.com	Male	1990027530526	2024-12-08 05:12:31
CIT004	Nur Ibne Kawsar Zitu	House 226, Road 5, Block C, Bashundhara R/A	01712345670	2000-11-15	nur.zitu@gmail.com	Male	1990045530425	2024-12-08 05:12:31
CIT005	Sabbir Hossain	House 126, Road 15, Block A, Bashundhara R/A	01798766832	2000-04-09	sabbirsts420@gmail.com	Male	1990025530223	2024-12-08 05:12:31
CIT006	Sayed Mahi Ashrafi	House 26, Road 2, Block D, Nikunja R/A	01798765432	2001-09-17	syed.mahi@gmail.com	Male	1990067530124	2024-12-08 05:12:31
CIT007	Ivy Khan	House 427, Road 8, Block D, Bashundhara R/A	01745678901	2001-08-06	ivy.khan@gmail.com	Female	1990024498128	2024-12-08 05:12:31
CIT008	Meher Easha	House 126, Road 8, Block C, Bashundhara R/A	01711223344	2001-08-15	emeherun@gmail.com	Female	1990024530129	2024-12-08 05:12:31
CIT009	Md. Tanvir Hossain	House 15, Road 10, Gulshan 1, Dhaka	01712345678	1999-10-26	tanvir.hossain@gmail.com	Male	1990024530123	2024-12-08 05:12:31
CIT010	Nasima Akhter	Flat 5C, Lakeview Apartments, Banani, Dhaka	01812345679	1999-05-18	nasima.akhter@yahoo.com	Female	1990054525678	2024-12-08 05:12:31
CIT011	Shahriar Kabir	House 24, Road 8, Dhanmondi, Dhaka	01912345670	1990-08-26	shahriar.kabir@gmail.com	Male	1998014530987	2024-12-08 05:12:31
CIT012	Ayesha Rahman	House 32, Sector 7, Uttara, Dhaka	01798765432	1996-01-26	ayesha.rahman@gmail.com	Female	1995056783456	2024-12-08 05:12:31
CIT013	Md. Abir Hasan	House 18, Road 5, Mohammadpur, Dhaka	01876543210	1987-10-06	abir.hasan@outlook.com	Male	1996034526789	2024-12-08 05:12:31
CIT014	Taslima Begum	Flat 4A, Green Plaza, Mirpur 12, Dhaka	01745678901	1975-10-26	taslima.begum@gmail.com	Female	1993034786541	2024-12-08 05:12:31
CIT015	Khaled Mahmud	House 22, Bashundhara R/A, Block F, Dhaka	01899887766	1985-12-16	khaled.mahmud@gmail.com	Male	1998012567894	2024-12-08 05:12:31
CIT017	Raiyan Ahmed	House 22, Banasree, Block C, Dhaka	014589887766	2010-12-16	raiyan.ahmed@gmail.com	Male	1998012567895	2024-12-08 05:12:31
CIT108	Reshman Nuriva	House 158 Dhanmondi-7, Dhaka	01405872035	2002-01-06	reshman.nuriva@outlook.com	Female	1998012567492	2024-12-10 04:57:54
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 6: Citizen Info Table

```sql
CREATE TABLE Citizen_Info (
    Citizen_ID VARCHAR(30) PRIMARY KEY,
    Name VARCHAR(200) NOT NULL,
    Address VARCHAR(300) NOT NULL,
    Contact_Number VARCHAR(15) NOT NULL,
    Date_Of_Birth DATE NOT NULL,
    Email VARCHAR(100) UNIQUE,
    Gender VARCHAR(10) CHECK (Gender IN ('Male', 'Female', 'Others')),
    National_ID VARCHAR(30) UNIQUE NOT NULL,
    City_Registration_Date TIMESTAMP DEFAULT CURRENT_TIMESTAMP NOT NULL
);
```

- **City Official:**

Official_ID	Citizen_ID	Name	Department_ID	Role	Phone_Number	Email	Years_Of_Service	Qualifications	Address	Supervisor_ID
CF1	CIT001	Professor Dr. Kamruddin Nur	1	City Mayor	01777354323	kamruddin.nur@citygov.org	15	PhD in Ubiquitous Computing	House: 91; Road:04; Block:B, Banani Dhaka-1213	NULL
DEF001	CIT002	Hasnat Karibul Islam	101	Department Head	011316619278	hasnat.islam@citygov.org	13	PhD in Molecular Engineering	House 526, Road 17, Block J, Bashundhara R/A	CF1
DEF002	CIT004	Nur Ibne Kawsar Zitun	101	Officer	01876543210	nur.zitu@citygov.org	3	MSc in Software Engineering	House 226, Road 5, Block C, Bashundhara R/A	DEF001
EDU001	CIT006	Sayed Mahi Ashrafi	108	Department Head	01711221122	mahi.ashrafi@citygov.org	12	PhD in Education Management	House 26, Road 2, Block D, Nikunja R/A	CF1
EDU002	CIT008	Meher Easha	108	Officer	01711223344	meher.easha@citygov.org	6	Masters in Public Administration	House 126, Road 8, Block C, Bashundhara R/A	EDU001
FIN001	CIT007	Ivy Khan	109	Department Head	01316619278	ivy.khan@citygov.org	10	PhD in Management AND HR	House 427, Road 8, Block D, Bashundhara R/A	CF1
FIN002	CIT012	Ayesha Rahman	109	Senior Officer	01712345678	ayesha.rahman@citygov.org	5	Masters in Urban Planning	House 32, Sector 7, Uttara, Dhaka	FIN001
HLT001	CIT003	Md. Minhajul Islam	107	Department Head	01756619289	minhajul.islam@citygov.org	12	PhD in Pharmaceutical Science	House 257, Road 5, Block C, Bashundhara R/A	CF1
HLT002	CIT015	Khaled Mahmud	107	Senior Officer	01899887766	khaled.mahmud@citygov.org	12	Bachelorâ€™s in Environmental Engineering	House 22, Bashundhara R/A, Block F, Dhaka	HLT001
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 7: City Official Table

```sql
CREATE TABLE City_Official (
    Official_ID VARCHAR(50) PRIMARY KEY,
    Citizen_ID VARCHAR(50),
    Name VARCHAR(100) NOT NULL,
    Department_ID VARCHAR(10) NOT NULL,
    Role VARCHAR(50) NOT NULL,
    Phone_Number VARCHAR(15),
    Email VARCHAR(100),
    Years_Of_Service INT,
    Qualifications TEXT,
    Address TEXT,
    Supervisor_ID VARCHAR(50),
    FOREIGN KEY (Citizen_ID) REFERENCES Citizen_Info(Citizen_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    FOREIGN KEY (Supervisor_ID) REFERENCES City_Official(Official_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

```

- **Department:**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Department_ID	Department_Name	Head_Official_ID	Department_Budget	Number_Of_Employees	Main_Responsibilities	Operating_Hours
1	Department of Head of the city	CF1	515000000.00	900	Head of the City and Policy makers.	24/7
101	Defence and Public Safety	DEF001	50000000.00	120	Manage law enforcement, fire services, and pu...	24/7
102	Infrastructure	CF1	80000000.00	150	Oversee construction, maintenance of roads, b...	9:00 AM - 5:00 PM
103	Transportation	CF1	60000000.00	100	Manage public transport systems, metro service...	6:00 AM - 11:00 PM
104	Environment and Sanitation	CF1	30000000.00	80	Ensure waste management, clean water supply...	8:00 AM - 6:00 PM
105	Urban Planning	CF1	45000000.00	60	Plan and regulate land use, zoning, and urban d...	9:00 AM - 5:00 PM
106	Emergency Services	CF1	70000000.00	90	Respond to emergencies, manage disaster reco...	24/7
107	Health and Welfare	HLT001	55000000.00	110	Provide health services, welfare programs, and ...	9:00 AM - 5:00 PM
108	Education and Community Services	EDU001	40000000.00	75	Oversee public schools, libraries, and communit...	8:00 AM - 6:00 PM
109	Finance and Budget	FIN001	20000000.00	50	Manage city finances, budgets, and tax collection.	9:00 AM - 5:00 PM
110	Technology and Innovation	CF1	65000000.00	40	Develop and manage city tech infrastructure an...	9:00 AM - 6:00 PM
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 8: Department Table

```sql
CREATE TABLE Department (
    Department_ID VARCHAR(10) PRIMARY KEY,
    Department_Name VARCHAR(100) NOT NULL,
    Head_Official_ID VARCHAR(50),
    Department_Budget DECIMAL(15, 2),
    Number_Of_Employees INT,
    Main_Responsibilities TEXT,
    Operating_Hours VARCHAR(50),
    FOREIGN KEY (Head_Official_ID) REFERENCES City_Official(Official_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

```

- **Public Services:**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Service_ID	Department_ID	Service_Type	Service_Description	Availability_Status	Average_Response_Time	Service_Fee
PS001	101	Emergency Rescue	Provide immediate rescue operations during nat...	Available	0.33	0.00
PS002	102	Bridge Inspection	Conduct routine inspections and maintenance o...	Available	72.00	0.00
PS003	PS002	Traffic Management	Monitor and regulate traffic flow across the city.	Available	0.25	0.00
PS004	104	Water Quality Monitoring	Test and ensure safe drinking water supply to h...	Unavailable	NULL	200.00
PS005	105	Urban Redevelopment	Plan and execute redevelopment projects in un...	Available	96.00	10000.00
PS006	106	Disaster Recovery	Coordinate recovery efforts after major disasters.	Available	2.00	0.00
PS007	107	Vaccination Programs	Administer free vaccines for common diseases.	Available	0.50	0.00
PS008	108	Library Membership	Provide access to public libraries and online reso...	Available	0.08	500.00
PS009	109	Tax Consultation	Assist citizens in calculating and paying city taxes.	Available	0.08	10000.00
PS010	110	Wi-Fi Services	Offer free public Wi-Fi in selected urban zones.	Available	0.17	500.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 9: Public Service Table

```sql
CREATE TABLE Public_Services (
    Service_ID VARCHAR(30) PRIMARY KEY,
    Department_ID VARCHAR(10),
    Service_Type VARCHAR(100) NOT NULL,
    Service_Description TEXT,
    Availability_Status VARCHAR(20) CHECK (Availability_Status IN ('Available',
    'Unavailable')),
    Average_Response_Time DECIMAL(5, 2),
    Service_Fee DECIMAL(10, 2) DEFAULT 0.00,
    FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

```

- **Feedback:**

Feedback_ID	Citizen_ID	Service_ID	Assigned_Official_ID	Feedback_Date	Feedback_Type	Description	Rating	Resolved_Status	Resolution_Date
FB001	CIT001	PS001	DEF002	2024-12-01	Complaint	The service was delayed.	2.0	0	NULL
FB002	CIT002	PS002	CF1	2024-12-02	Praise	The officer was very helpful and efficient.	5.0	1	2024-12-02
FB003	CIT003	PS003	CF1	2024-12-03	Suggestion	Consider extending service hours to weekends.	4.0	0	NULL
FB004	CIT004	PS004	CF1	2024-12-04	Complaint	Difficulties in reaching support during emergenci...	3.0	0	NULL
FB005	CIT005	PS005	CF1	2024-12-05	Praise	Prompt response and excellent resolution.	5.0	1	2024-12-05
FB006	CIT006	PS006	CF1	2024-12-06	Complaint	Response time could be improved.	3.5	0	NULL
FB007	CIT007	PS007	HLT002	2024-12-07	Praise	Vaccination service was very organized.	5.0	1	2024-12-07
FB008	CIT008	PS008	EDU002	2024-12-08	Suggestion	Consider reducing the membership fee.	4.5	0	NULL
FB009	CIT009	PS009	FIN002	2024-12-09	Complaint	Tax consultation fee is too high.	2.0	0	NULL
FB010	CIT010	PS010	CF1	2024-12-10	Praise	Wi-Fi service is fast and reliable.	5.0	1	2024-12-10
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 10: Feedback Table

```sql
CREATE TABLE Feedback (
    Feedback_ID VARCHAR(10) PRIMARY KEY,
    Citizen_ID VARCHAR(50),
    Service_ID VARCHAR(30),
    Assigned_Official_ID VARCHAR(50),
    Feedback_Date DATE NOT NULL,
    Feedback_Type VARCHAR(50),
    Description TEXT,
    Rating DECIMAL(2, 1) CHECK (Rating BETWEEN 1 AND 5),
    Resolved_Status BOOLEAN DEFAULT FALSE,
    Resolution_Date DATE DEFAULT NULL,
    FOREIGN KEY (Citizen_ID) REFERENCES Citizen_Info(Citizen_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    FOREIGN KEY (Service_ID) REFERENCES Public_Services(Service_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    FOREIGN KEY (Assigned_Official_ID) REFERENCES City_Official(Official_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

- **Service Request:**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Request_ID	Citizen_ID	Service_ID	Request_Date	Request_Status	Assigned_Official_ID	Request_Description	Completion_Date
REQ001	CIT005	PS001	2024-12-01	Pending	CF1	Need immediate rescue operations after flood d...	NULL
REQ002	CIT015	PS002	2024-12-02	In Progress	DEF002	Routine inspection for bridge maintenance.	NULL
REQ003	CIT014	PS003	2024-12-03	Completed	CF1	Traffic signal repair in downtown area.	2024-12-05
REQ004	CIT012	PS004	2024-12-04	Rejected	CF1	Water quality testing for municipal supply.	NULL
REQ005	CIT005	PS005	2024-12-05	Pending	CF1	Urban redevelopment planning for old city blocks.	NULL
REQ006	CIT006	PS006	2024-12-06	In Progress	CF1	Assistance required for disaster recovery.	NULL
REQ007	CIT007	PS007	2024-12-07	Completed	HLT002	Vaccination program support for school children.	2024-12-08
REQ008	CIT008	PS008	2024-12-08	Pending	EDU002	Library membership application.	NULL
REQ009	CIT009	PS009	2024-12-09	Rejected	FIN002	Tax consultation regarding new property.	NULL
REQ010	CIT010	PS010	2024-12-10	Completed	CF1	Wi-Fi service setup in urban zone.	2024-12-12
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 11: Service Request Table

```sql
CREATE TABLE Service_Request (
    Request_ID VARCHAR(50) PRIMARY KEY,
    Citizen_ID VARCHAR(50) NOT NULL,
    Service_ID VARCHAR(30),
    Request_Date DATE NOT NULL,
    Request_Status VARCHAR(20) DEFAULT 'Pending'
        CHECK (Request_Status IN ('Pending', 'In Progress', 'Completed', '
    Rejected')),
    Assigned_Official_ID VARCHAR(50),
    Request_Description TEXT,
    Completion_Date DATE,
    FOREIGN KEY (Citizen_ID) REFERENCES Citizen_Info(Citizen_ID)
        ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (Service_ID) REFERENCES Public_Services(Service_ID)
        ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (Assigned_Official_ID) REFERENCES City_Official(Official_ID)
        ON DELETE SET NULL ON UPDATE CASCADE
);
```

- **Location:**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Location_ID	Location_Name	Location_Type	City	Region	Postal_Code	Additional_Info
DHAKA0	Paltan	Commercial	Dhaka	Dhaka South	1000	Known for political rallies and trade centers.
DHAKA1	Gulshan	Residential	Dhaka	Dhaka North	1212	High-end residential area with embassies.
DHAKA11	Bashundhara Residential Area	Mixed-Use	Dhaka	Dhaka North	1229	Developed area with residential and commercial ...
DHAKA12	Tejgaon	Industrial	Dhaka	Dhaka North	1215	Prominent industrial area with TV stations.
DHAKA13	Farmgate	Mixed-Use	Dhaka	Dhaka North	1215	Busy commercial hub with schools and offices.
DHAKA14	Shyamoli	Residential	Dhaka	Dhaka North	1207	Area with parks and affordable housing.
DHAKA15	Keraniganj	Industrial	Dhaka	Dhaka South	1310	Known for the garment industry and factories.
DHAKA16	Jatrabari	Commercial	Dhaka	Dhaka South	1204	Busy area known for transportation hubs.
DHAKA17	Khilgaon	Mixed-Use	Dhaka	Dhaka South	1219	Residential area with growing commercial zones.
DHAKA18	Rampura	Residential	Dhaka	Dhaka South	1219	Area with television studios and middle-class ho...
DHAKA19	Savar	Industrial	Dhaka	Dhaka North	1340	Home to EPZ and garment industries.
DHAKA2	Banani	Mixed-Use	Dhaka	Dhaka North	1213	Popular for both residential and commercial spa...
DHAKA20	Aminbazar	Mixed-Use	Dhaka	Dhaka North	1348	Growing urban area near Savar.
DHAKA21	Kamalapur	Commercial	Dhaka	Dhaka South	1223	Location of the central railway station of Dhaka.
DHAKA22	Green Road	Residential	Dhaka	Dhaka South	1205	Popular area for middle-class residences and me...
DHAKA23	Malibagh	Mixed-Use	Dhaka	Dhaka South	1217	Busy area with shopping malls and apartments.
DHAKA24	Lalbagh	Heritage	Dhaka	Dhaka South	1211	Famous for Lalbagh Fort, a Mughal-era monume...
DHAKA25	Hatirjheel	Mixed-Use	Dhaka	Dhaka North	1212	Urban development project with recreational sp...
DHAKA26	Khilkhet	Residential	Dhaka	Dhaka North	1229	A growing suburban area near Bashundhara.
DHAKA27	Turag	Residential	Dhaka	Dhaka North	1711	Located near the river, with increasing residenti...
DHAKA28	Banasree	Residential	Dhaka	Dhaka South	1219	Planned housing area ideal for families.
DHAKA29	Agargaon	Mixed-Use	Dhaka	Dhaka North	1207	Home to government offices and IT parks.
DHAKA3	Dhanmondi	Residential	Dhaka	Dhaka South	1209	Known for educational institutions and lakes.
DHAKA30	Ashulia	Industrial	Dhaka	Dhaka North	1341	Key industrial hub for garments and manufactur...
DHAKA4	Motijheel	Commercial	Dhaka	Dhaka South	1000	Central business district of Dhaka.
DHAKA5	Old Dhaka	Heritage	Dhaka	Dhaka South	1100	Historic area with narrow streets and Mughal-er...
DHAKA6	Uttara	Residential	Dhaka	Dhaka North	1230	A planned residential area near the airport.
DHAKA7	Mirpur	Residential	Dhaka	Dhaka North	1216	Famous for the National Zoo and Sher-e-Bangla...
DHAKA8	Baridhara	Residential	Dhaka	Dhaka North	1212	Diplomatic area housing embassies and high-en...
DHAKA9	Mohakhali	Commercial	Dhaka	Dhaka North	1212	Hub for hospitals and pharmaceutical companies.
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 12: Location Table

```sql
CREATE TABLE Location (
    Location_ID VARCHAR(50) PRIMARY KEY,
    Location_Name VARCHAR(100) NOT NULL,
    Location_Type VARCHAR(50) CHECK (Location_Type IN ('Residential', '
    Commercial', 'Industrial', 'Mixed-Use', 'Heritage')),
    City VARCHAR(50) NOT NULL,
    Region VARCHAR(50) NOT NULL,
    Postal_Code VARCHAR(10),
    Additional_Info TEXT
);
```

- **Citizen Location:**

Citizen_ID	Location_ID
CIT009	DHAKA1
CIT002	DHAKA11
CIT003	DHAKA11
CIT004	DHAKA11
CIT005	DHAKA11
CIT007	DHAKA11
CIT008	DHAKA11
CIT015	DHAKA11
CIT010	DHAKA12
CIT013	DHAKA14
CIT001	DHAKA2
CIT006	DHAKA26
CIT011	DHAKA3
CIT012	DHAKA6
CIT014	DHAKA7
NULL	NULL

Figure 13: Citizen Location Table

```sql
CREATE TABLE Citizen_Location (
    Citizen_ID VARCHAR(30),
    Location_ID VARCHAR(50),
    PRIMARY KEY (Citizen_ID, Location_ID),
    FOREIGN KEY (Citizen_ID) REFERENCES Citizen_Info(Citizen_ID) ON DELETE
   CASCADE,
    FOREIGN KEY (Location_ID) REFERENCES Location(Location_ID) ON DELETE CASCADE
);

```

- **Weather Condition:**

Weather_ID	Location_ID	Observation_Date	Temperature	Humidity	Weather_Condition	Wind_Speed	Additional_Info
W001	DHAKA1	2024-12-01	25.50	60	Sunny	15.20	Clear sky with mild breeze
W002	DHAKA2	2024-12-01	22.30	80	Rainy	10.50	Light rain in the morning
W003	DHAKA3	2024-12-01	30.10	50	Cloudy	12.40	Overcast skies throughout the day
W004	DHAKA4	2024-12-01	18.70	85	Stormy	45.00	Heavy rain and strong winds
W005	DHAKA5	2024-12-01	15.50	90	Foggy	5.00	Low visibility in the early morning
W006	DHAKA6	2024-12-01	28.80	70	Windy	25.80	Strong gusts in the afternoon
W007	DHAKA7	2024-12-01	12.30	95	Foggy	8.20	Light rainfall overnight
W008	DHAKA8	2024-12-01	29.90	55	Sunny	18.50	Warm day with a clear sky
W009	DHAKA9	2024-12-01	21.40	75	Rainy	9.30	Drizzle during the evening
W010	DHAKA0	2024-12-01	26.00	65	Cloudy	11.00	Partly cloudy with no precipitation
W011	DHAKA11	2024-12-02	24.80	68	Sunny	14.20	Moderate temperatures with clear...
W012	DHAKA12	2024-12-02	20.50	85	Rainy	8.70	Rain showers expected in the aft...
W013	DHAKA13	2024-12-02	31.00	45	Cloudy	13.50	Hot day with occasional clouds
W014	DHAKA14	2024-12-02	17.00	90	Foggy	6.00	Dense fog in the early hours
W015	DHAKA15	2024-12-02	27.50	60	Windy	22.40	Breezy conditions throughout the ...
W016	DHAKA16	2024-12-02	16.50	95	Cloudy	10.00	Rainfall with high humidity
W017	DHAKA17	2024-12-02	29.20	58	Sunny	16.80	Clear day with moderate wind
W018	DHAKA18	2024-12-02	22.70	80	Rainy	7.60	Intermittent rain with cool temper...
W019	DHAKA19	2024-12-02	25.30	70	Cloudy	12.00	Cloudy skies with mild humidity
W020	DHAKA20	2024-12-02	30.50	50	Sunny	20.30	Hot and sunny day with clear skies
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 14: Weather Condition Table

```sql
CREATE TABLE Weather_Condition_Data (
    Weather_ID VARCHAR(50) PRIMARY KEY,
    Location_ID VARCHAR(50) NOT NULL,
    Observation_Date DATE NOT NULL,
    Temperature DECIMAL(5, 2) NOT NULL,
    Humidity INT CHECK (Humidity BETWEEN 0 AND 100),
    Weather_Condition VARCHAR(50) CHECK (
        Weather_Condition IN ('Sunny', 'Rainy', 'Cloudy', 'Stormy', 'Windy', '
    Foggy')),
    Wind_Speed DECIMAL(5, 2),
    Additional_Info TEXT,
    FOREIGN KEY (Location_ID) REFERENCES Location(Location_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

- **Infrastructure Maintenance:**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Maintenance_ID	Location_ID	Department_ID	Maintenance_Type	Start_Date	End_Date	Status	Cost	Contractor_Name	Additional_Info
M001	DHAKA1	103	Gulshan Link Road Repair	2024-01-01	2024-01-10	Completed	15000.00	Abdul Momen Contractors	Repair of potholes on Main Street.
M002	DHAKA2	109	Bank Building Maintainance	2024-12-15	NULL	Ongoing	20000.00	24th Army Brigade	Upgrading and maitainance of the diffected ban...
M004	DHAKA11	102	Bridge Inspection	2024-03-01	2024-03-05	Completed	5000.00	BridgeCo Inspections	Annual safety inspection of the River Bridge.
M005	DHAKA9	109	Hospital building repair	2024-03-10	NULL	Planned	3000.00	Abdul Momen Group	ICDDRB building repair.
M006	DHAKA9	NULL	Park Renovation	2024-01-20	2024-02-20	Completed	25000.00	Green Earth Landscaping	Renovation of the central park with new pathw...
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 15: Infrastructure Maintenance Table

```sql
CREATE TABLE Infrastructure_Maintenance (
    Maintenance_ID VARCHAR(50) PRIMARY KEY,
    Location_ID VARCHAR(50) NOT NULL,
    Department_ID VARCHAR(50),
    Maintenance_Type VARCHAR(100) NOT NULL,
    Start_Date DATE NOT NULL,
    End_Date DATE,
    Status VARCHAR(50) CHECK (Status IN ('Planned', 'Ongoing', 'Completed'))
   DEFAULT 'Planned',
    Cost DECIMAL(15,2),
    Contractor_Name VARCHAR(100),
    Additional_Info TEXT,
    FOREIGN KEY (Location_ID) REFERENCES Location(Location_ID) ON DELETE CASCADE
     ON UPDATE CASCADE,
    FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

- **Bus Transportation:**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Bus_ID	Route_Name	Start_Location_ID	End_Location_ID	Number_Of_Stations	Frequency_Per_Hour	status
Anabil Paribahan	Route H	DHAKA8	DHAKA17	14	6	On Time
Ashin Paribahan	Route D	DHAKA4	DHAKA13	18	7	Cancelled
Azam Paribahan	Route J	DHAKA0	DHAKA19	16	5	On Time
Dhakar Chaka	Route E	DHAKA5	DHAKA14	20	8	On Time
Green Dhaka	Route A	DHAKA1	DHAKA0	15	6	On Time
Projapoti Paribahan	Route G	DHAKA7	DHAKA16	25	10	On Time
Retro Paribahan	Route I	DHAKA9	DHAKA18	9	4	Cancelled
Sikhor Paribhan	Route C	DHAKA3	DHAKA12	10	4	On Time
Torongo Paribahan	Route B	DHAKA2	DHAKA11	12	5	Delayed
Victor Classic	Route F	DHAKA6	DHAKA15	8	3	Delayed
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 16: Bus Transportation Table

```sql
CREATE TABLE Bus_Transportation (
    Bus_ID VARCHAR(30) PRIMARY KEY,
    Route_Name VARCHAR(50) NOT NULL,
    Start_Location_ID VARCHAR(50),
    End_Location_ID VARCHAR(50),
    Number_Of_Stations INT NOT NULL,
    Frequency_Per_Hour INT NOT NULL,
    status ENUM('On Time', 'Delayed', 'Cancelled') DEFAULT 'On Time',
    FOREIGN KEY (Start_Location_ID) REFERENCES Location(Location_ID) ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (End_Location_ID) REFERENCES Location(Location_ID) ON DELETE CASCADE ON UPDATE CASCADE
);
```

- **IoT Devices:**

Device_ID	Device_Type	Department_ID	Location_ID	Installation_Date	Device_Status	Last_Communication_Timestamp	Manufacturer	Battery_Status
SENSOR001	Temperature Sensor	104	DHAKA1	2023-01-15	Active	2024-12-07 14:20:00	SensTech	85%
SENSOR002	Humidity Sensor	104	DHAKA2	2023-02-20	Active	2024-12-07 14:25:00	WeatherPro	78%
SENSOR003	Air Quality Monitor	104	DHAKA3	2023-03-10	Inactive	2024-12-07 13:50:00	EnviroCheck	Battery Low
SENSOR004	Water Level Sensor	104	DHAKA4	2023-04-05	Active	2024-12-07 14:30:00	AquaSense	92%
SENSOR005	Motion Detector	110	DHAKA5	2023-05-12	Active	2024-12-07 14:15:00	SecureIt	80%
SENSOR006	Pressure Sensor	110	DHAKA6	2023-06-08	Active	2024-12-07 14:10:00	PressurTech	75%
SENSOR007	Proximity Sensor	110	DHAKA7	2023-07-14	Inactive	2024-12-06 18:45:00	ProxiMatic	60%
SENSOR008	Light Intensity Sensor	110	DHAKA8	2023-08-01	Active	2024-12-07 14:00:00	BrightSense	90%
SENSOR009	Sound Level Sensor	110	DHAKA9	2023-09-19	Active	2024-12-07 13:55:00	AcoustiCare	82%
SENSOR010	Gas Leak Detector	101	DHAKA0	2023-10-25	Active	2024-12-07 14:05:00	SafeGasTech	88%
SENSOR011	Infrared Sensor	104	DHAKA11	2023-11-11	Active	2024-12-07 14:40:00	IRMax	95%
SENSOR012	UV Index Sensor	110	DHAKA12	2023-12-02	Inactive	2024-12-06 15:20:00	UltraSafe	Battery Low
SENSOR013	Vibration Sensor	110	DHAKA13	2023-01-28	Active	2024-12-07 14:50:00	VibraTech	70%
SENSOR014	Smoke Detector	104	DHAKA14	2023-02-16	Inactive	2024-12-06 17:30:00	FireSecure	Low Battery
SENSOR015	Flood Sensor	104	DHAKA15	2023-03-05	Active	2024-12-07 14:35:00	FloodGuard	85%
SENSOR016	GPS Tracker	110	DHAKA16	2023-04-20	Active	2024-12-07 14:45:00	GeoLocator	77%
SENSOR017	Barcode Scanner	110	DHAKA17	2023-05-30	Active	2024-12-07 13:50:00	ScanTech	90%
SENSOR018	RFID Sensor	110	DHAKA18	2023-06-25	Active	2024-12-07 14:55:00	IDSecure	Battery Full
SENSOR019	Magnetometer	110	DHAKA19	2023-07-10	Active	2024-12-07 13:45:00	MagSense	85%
SENSOR020	Energy Meter	110	DHAKA20	2023-08-15	Active	2024-12-07 14:10:00	EnerTech	88%
SENSOR021	Traffic Sensor	103	DHAKA1	2023-01-15	Active	2024-11-01 08:00:00	SensorTech	80%
SENSOR022	Traffic Sensor	103	DHAKA2	2023-02-10	Active	2024-11-01 09:00:00	SensorTech	65%
SENSOR023	Traffic Sensor	103	DHAKA3	2023-03-05	Active	2024-11-01 07:45:00	SensorTech	62%
SENSOR024	Traffic Sensor	103	DHAKA4	2023-04-20	Active	2024-11-01 10:15:00	SensorTech	68%
SENSOR025	Traffic Sensor	103	DHAKA5	2023-05-10	Active	2024-11-01 11:00:00	SensorTech	54%
SENSOR026	Traffic Sensor	103	DHAKA6	2023-06-15	Active	2024-11-01 06:30:00	SensorTech	59%
SENSOR027	Traffic Sensor	103	DHAKA7	2023-07-25	Active	2024-11-01 17:00:00	SensorTech	5%
SENSOR028	Traffic Sensor	103	DHAKA8	2023-08-30	Active	2024-11-01 12:30:00	SensorTech	45%
SENSOR029	Traffic Sensor	103	DHAKA8	2023-09-15	Active	2024-11-01 18:45:00	SensorTech	95%
SENSOR030	Traffic Sensor	103	DHAKA9	2023-10-10	Active	2024-11-01 05:30:00	SensorTech	100%
SENSOR031	Traffic Sensor	103	DHAKA0	2023-01-15	Active	2024-11-01 08:00:00	SensorTech	75%
SENSOR032	Traffic Sensor	103	DHAKA11	2023-02-10	Active	2024-11-01 09:00:00	SensorTech	49%
SENSOR033	Traffic Sensor	103	DHAKA12	2023-03-05	Active	2024-11-01 07:45:00	SensorTech	69%
SENSOR034	Traffic Sensor	103	DHAKA13	2023-04-20	Active	2024-11-01 10:15:00	SensorTech	90%
SENSOR035	Traffic Sensor	103	DHAKA14	2023-05-10	Active	2024-11-01 11:00:00	SensorTech	58%
SENSOR036	Traffic Sensor	103	DHAKA15	2023-06-15	Active	2024-11-01 06:30:00	SensorTech	64%
SENSOR037	Traffic Sensor	103	DHAKA16	2023-07-25	Active	2024-11-01 17:00:00	SensorTech	45%

Figure 17: IoT Devices Table

```sql
CREATE TABLE IoT_Devices (
    Device_ID VARCHAR(30) PRIMARY KEY,
    Device_Type VARCHAR(100) NOT NULL,
    Department_ID VARCHAR(10) NOT NULL,
    Location_ID VARCHAR(50) NOT NULL,
    Installation_Date DATE NOT NULL,
    Device_Status VARCHAR(50) NOT NULL,
    Last_Communication_Timestamp DATETIME NOT NULL,
    Manufacturer VARCHAR(100),
    Battery_Status VARCHAR(50),
    FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID),
    FOREIGN KEY (Location_ID) REFERENCES Location(Location_ID)
);

```

- **Traffic Data:**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Traffic_ID	Location_ID	Device_ID	Vehicle_Count	Average_Speed	Accident_Reports	Congestion_Level	Timestamp
T001	DHAKA1	SENSOR021	120	45.50	1	Moderate	2024-12-07 15:00:00
T002	DHAKA2	SENSOR022	200	40.30	0	High	2024-12-07 15:05:00
T003	DHAKA3	SENSOR023	50	55.20	0	Low	2024-12-07 15:10:00
T004	DHAKA4	SENSOR024	80	35.10	2	High	2024-12-07 15:15:00
T005	DHAKA5	SENSOR025	150	42.80	0	Moderate	2024-12-07 15:20:00
T006	DHAKA6	SENSOR026	90	48.60	1	Moderate	2024-12-07 15:25:00
T007	DHAKA7	SENSOR027	110	50.40	0	Moderate	2024-12-07 15:30:00
T008	DHAKA8	SENSOR028	180	38.70	1	High	2024-12-07 15:35:00
T009	DHAKA9	SENSOR029	70	60.10	0	Low	2024-12-07 15:40:00
T010	DHAKA0	SENSOR030	130	43.90	3	High	2024-12-07 15:45:00
T011	DHAKA11	SENSOR031	100	47.30	0	Moderate	2024-12-07 15:50:00
T012	DHAKA12	SENSOR032	60	53.40	1	Low	2024-12-07 15:55:00
T013	DHAKA13	SENSOR033	140	41.80	2	High	2024-12-07 16:00:00
T014	DHAKA14	SENSOR034	220	35.60	3	High	2024-12-07 16:05:00
T015	DHAKA15	SENSOR035	90	50.90	0	Moderate	2024-12-07 16:10:00
T016	DHAKA16	SENSOR036	180	42.30	0	Moderate	2024-12-07 16:15:00
T017	DHAKA17	SENSOR037	120	48.70	1	Moderate	2024-12-07 16:20:00
T018	DHAKA18	SENSOR038	110	49.10	0	Moderate	2024-12-07 16:25:00
T019	DHAKA19	SENSOR039	160	40.20	2	High	2024-12-07 16:30:00
T020	DHAKA20	SENSOR040	130	44.60	0	Moderate	2024-12-07 16:35:00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 18: Traffic Data Table

```sql
CREATE TABLE Traffic_Data (
    Traffic_ID VARCHAR(50) PRIMARY KEY,
    Location_ID VARCHAR(50) NOT NULL,
    Device_ID VARCHAR(30) NOT NULL,
    Vehicle_Count INT,
    Average_Speed DECIMAL(5, 2),
    Accident_Reports INT,
    Congestion_Level VARCHAR(50),
    Timestamp DATETIME NOT NULL,
    FOREIGN KEY (Location_ID) REFERENCES Location(Location_ID),
    FOREIGN KEY (Device_ID) REFERENCES IoT_Devices(Device_ID)
);

```

- **Utility Monitoring:**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Utility_ID	Location_ID	Utility_Type	Consumption	Provider	Timestamp	Cost_Per_Unit	Peak_Consumption_Period	Device_ID
U001	DHAKA1	Electricity	1200.50	PowerGrid Ltd.	2024-12-07 16:00:00	5.20	08:00-10:00	SENSOR001
U002	DHAKA2	Water	800.00	WaterSupply Co.	2024-12-07 16:05:00	2.10	06:00-08:00	SENSOR002
U003	DHAKA3	Gas	350.75	GasDistribution Inc.	2024-12-07 16:10:00	3.50	18:00-20:00	SENSOR003
U004	DHAKA4	Electricity	950.25	PowerGrid Ltd.	2024-12-07 16:15:00	5.20	09:00-11:00	SENSOR004
U005	DHAKA5	Water	1050.60	WaterSupply Co.	2024-12-07 16:20:00	2.10	07:00-09:00	SENSOR005
U006	DHAKA6	Gas	200.45	GasDistribution Inc.	2024-12-07 16:25:00	3.50	19:00-21:00	SENSOR006
U007	DHAKA7	Electricity	1150.80	PowerGrid Ltd.	2024-12-07 16:30:00	5.20	10:00-12:00	SENSOR007
U008	DHAKA8	Water	900.25	WaterSupply Co.	2024-12-07 16:35:00	2.10	05:00-07:00	SENSOR008
U009	DHAKA9	Gas	450.50	GasDistribution Inc.	2024-12-07 16:40:00	3.50	15:00-17:00	SENSOR009
U010	DHAKA0	Electricity	1300.00	PowerGrid Ltd.	2024-12-07 16:45:00	5.20	11:00-13:00	SENSOR010
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 19: Utility Monitoring Table

```sql
CREATE TABLE Utility_Monitoring (
    Utility_ID VARCHAR(50) PRIMARY KEY,
    Location_ID VARCHAR(30) NOT NULL,
    Utility_Type VARCHAR(50) NOT NULL,
    Consumption DECIMAL(10, 2) NOT NULL,
    Provider VARCHAR(100) NOT NULL,
    Timestamp DATETIME NOT NULL,
    Cost_Per_Unit DECIMAL(10, 2) NOT NULL,
    Peak_Consumption_Period VARCHAR(50),
    Device_ID VARCHAR(30) NOT NULL,
    FOREIGN KEY (Location_ID) REFERENCES Location(Location_ID),
    FOREIGN KEY (Device_ID) REFERENCES IoT_Devices(Device_ID)
);

```

7 Queries

7.1 Trigger

```
DELIMITER $$

CREATE TRIGGER After_Approval_Update
AFTER UPDATE ON CitizenRegistration
FOR EACH ROW
BEGIN
    IF NEW.Approval_Status = 'Approved' THEN
        INSERT INTO Citizen_Info (
            Citizen_ID, Name, Address, Contact_Number, Date_Of_Birth, Email, Gender,
    National_ID, City_Registration_Date
        )
        VALUES (
            NEW.Registration_ID, NEW.Full_Name, NEW.Address, NEW.Contact_Number,
            NEW.Date_Of_Birth, NEW.Email, NEW.Gender, NEW.National_ID, NEW.
    Registration_Date
        )
        ON DUPLICATE KEY UPDATE
            Name = VALUES(Name),
            Address = VALUES(Address),
            Contact_Number = VALUES(Contact_Number),
            Email = VALUES(Email),
            Gender = VALUES(Gender),
            National_ID = VALUES(National_ID);
    END IF;
END$$

DELIMITER ;

```

7.2 Adding the Foreign Key Constraint

```
ALTER TABLE City_Official
ADD CONSTRAINT FK_Department_ID
FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)
    ON DELETE CASCADE
    ON UPDATE CASCADE;

```

7.3 Citizen Registration Approval

```
UPDATE CitizenRegistration
SET Approval_Status = 'Approved'
WHERE Registration_ID LIKE 'CIT%' AND National_ID LIKE '199%';

```

7.4 Citizen Registration Rejection

```
UPDATE CitizenRegistration
SET Approval_Status = 'Rejected'
WHERE Registration_ID LIKE 'CIT%' AND National_ID NOT LIKE '199%';

```

7.5 Registration Applicants with Pending Status

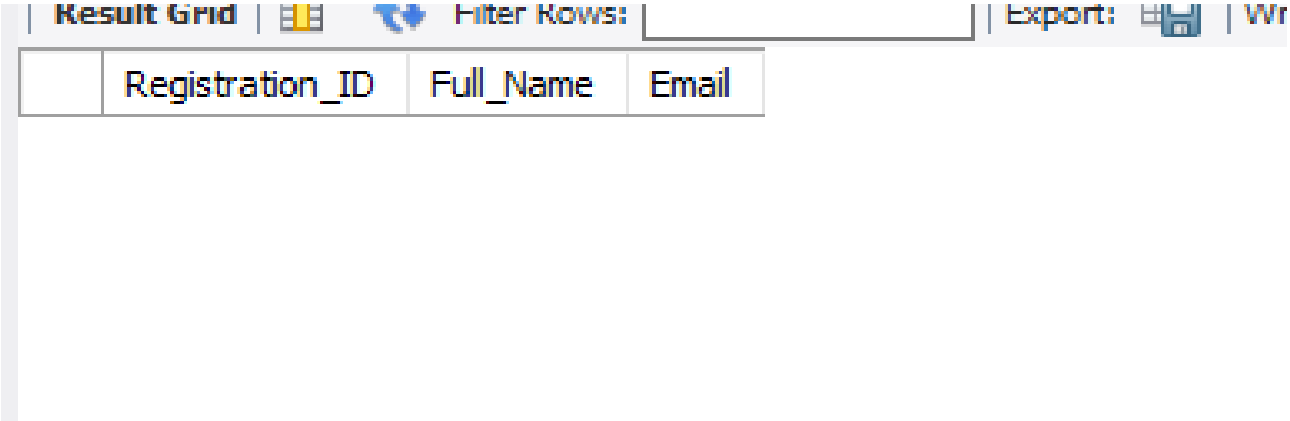


Figure 20: Pending status applicant Table

```
SELECT CR.Registration_ID, CR.Full_Name, CR.Email
FROM CitizenRegistration CR
LEFT JOIN Citizen_Info CI
ON CR.Registration_ID = CI.Citizen_ID
WHERE CI.Citizen_ID IS NULL AND CR.Approval_Status = 'Pending';

```

7.6 Citizens Registered This Month

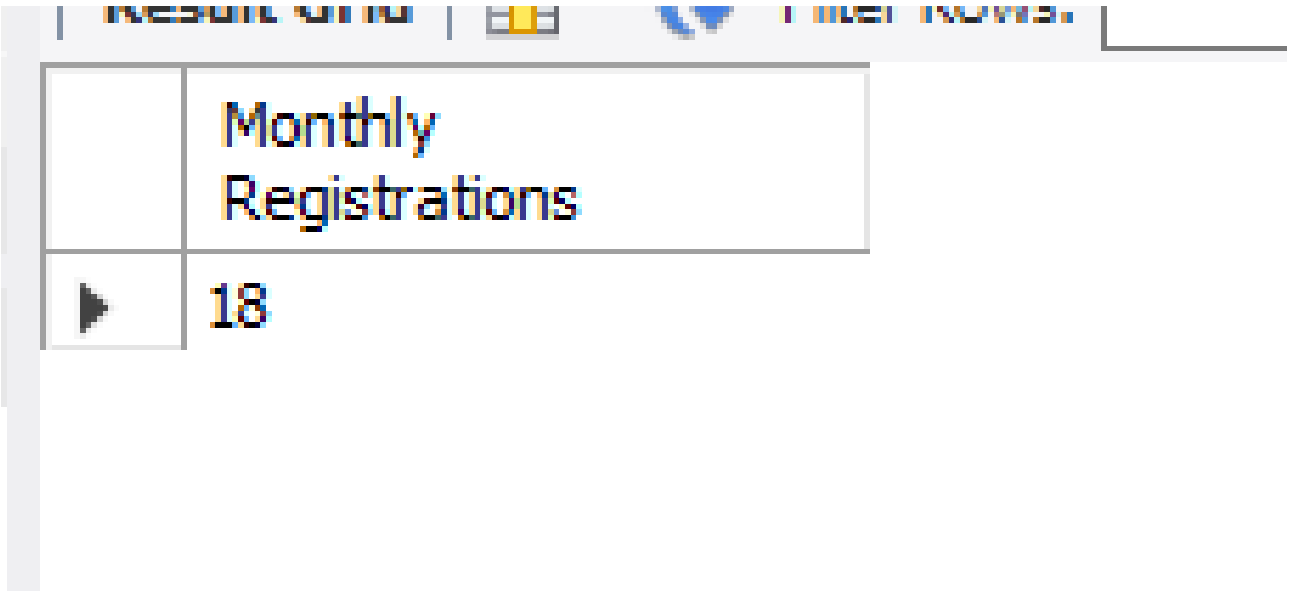


Figure 21: Citizens Registered This Month Table

```sql
SELECT COUNT(*) AS "Monthly Registrations"
FROM CitizenRegistration
WHERE MONTH(Registration_Date) = MONTH(CURDATE())
  AND YEAR(Registration_Date) = YEAR(CURDATE());

```

7.7 All Department Name and ID

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Department Name	Department Id
Department of Head of the city	1
Defence and Public Safety	101
Infrastructure	102
Transportation	103
Environment and Sanitation	104
Urban Planning	105
Emergency Services	106
Health and Welfare	107
Education and Community Services	108
Finance and Budget	109
Technology and Innovation	110

Figure 22: Department Name and ID Table

```
SELECT d.department_name AS "Department Name", d.Department_ID AS "Department Id"
FROM Department d;

```

7.8 Top Performing Departments in IoT Usage

Result Grid | Filter Rows: | Export: | Wrap

Department Name	Number of Active Devices
Transportation	22
Technology and Innovation	10
Environment and Sanitation	5
Defence and Public Safety	1

Figure 23: Top performing departments in IoT usage

```
SELECT d.Department_Name AS "Department Name", COUNT(iot.Device_ID) AS "Active
    Devices Number"
FROM IoT_Devices iot
JOIN Department d ON iot.Department_ID = d.Department_ID
WHERE iot.Device_Status = 'Active'
GROUP BY d.Department_ID
ORDER BY COUNT(iot.Device_ID) DESC;

```

7.9 Service Provided by Each Department

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Location	Location Id	Traffic Id	Congestion Level	Time
Paltan	DHAKA0	T020	Moderate	2024-12-07 16:35:00
Shyamoli	DHAKA14	T020	Moderate	2024-12-07 16:35:00
Jatrabari	DHAKA16	T020	Moderate	2024-12-07 16:35:00
Farmgate	DHAKA13	T020	Moderate	2024-12-07 16:35:00
Bashundhara Residential Area	DHAKA11	T020	Moderate	2024-12-07 16:35:00
Rampura	DHAKA18	T020	Moderate	2024-12-07 16:35:00
Tejgaon	DHAKA12	T020	Moderate	2024-12-07 16:35:00
Keraniganj	DHAKA15	T020	Moderate	2024-12-07 16:35:00
Khilgaon	DHAKA17	T020	Moderate	2024-12-07 16:35:00
Gulshan	DHAKA1	T020	Moderate	2024-12-07 16:35:00

Figure 24: Service and Department Table

```
SELECT d.Department_Name AS "Department Name", ps.Service_Type AS "Service Type"
FROM Public_Services ps
JOIN Department d ON ps.Department_ID = d.Department_ID;

```

7.10 Department Head

Department_Name	Head_Name
Department of Head of the city	Professor Dr. Kamruddin Nur
Defence and Public Safety	Hasnat Karibul Islam
Infrastructure	Professor Dr. Kamruddin Nur
Transportation	Professor Dr. Kamruddin Nur
Environment and Sanitation	Professor Dr. Kamruddin Nur
Urban Planning	Professor Dr. Kamruddin Nur
Emergency Services	Professor Dr. Kamruddin Nur
Health and Welfare	Md. Minhajul Islam
Education and Community Services	Sayed Mahi Ashrafi
Finance and Budget	Ivy Khan
Technology and Innovation	Professor Dr. Kamruddin Nur

Figure 25: Department Head Table

```
SELECT D.Department_Name, CO.Name AS Head_Name
FROM Department D
JOIN City_Official CO
ON D.Head_Official_ID = CO.Official_ID;

```

7.11 List of Services with Feedbacks

Service_Type	Feedback_Type	Rating
Emergency Rescue	Complaint	2.0
Bridge Inspection	Praise	5.0
Traffic Management	Suggestion	4.0
Water Quality Monitoring	Complaint	3.0
Urban Redevelopment	Praise	5.0
Disaster Recovery	Complaint	3.5
Vaccination Programs	Praise	5.0
Library Membership	Suggestion	4.5
Tax Consultation	Complaint	2.0
Wi-Fi Services	Praise	5.0

Figure 26: Services with Feedbacks Table

```sql
SELECT PS.Service_Type, F.Feedback_Type, F.Rating
FROM Public_Services PS
JOIN Feedback F
ON PS.Service_ID = F.Service_ID;

```

7.12 Counting Feedbacks by Services

Service_Type	Total_Feedbacks
Emergency Rescue	1
Bridge Inspection	1
Traffic Management	1
Water Quality Monitoring	1
Urban Redevelopment	1
Disaster Recovery	1
Vaccination Programs	1
Library Membership	1
Tax Consultation	1
Wi-Fi Services	1

Figure 27: Counting Feedbacks by Services Table

```sql
SELECT PS.Service_Type, COUNT(F.Feedback_ID) AS Total_Feedbacks
FROM Public_Services PS
JOIN Feedback F
ON PS.Service_ID = F.Service_ID
GROUP BY PS.Service_Type;

```

7.13 List Traffic Data by IoT Device

Result Grid | Filter Rows: | Export: | Wrap Cell C

Traffic_ID	Vehicle_Count	Device_Type
T001	120	Traffic Sensor
T002	200	Traffic Sensor
T003	50	Traffic Sensor
T004	80	Traffic Sensor
T005	150	Traffic Sensor
T006	90	Traffic Sensor
T007	110	Traffic Sensor
T008	180	Traffic Sensor
T009	70	Traffic Sensor
T010	130	Traffic Sensor
T011	100	Traffic Sensor
T012	60	Traffic Sensor
T013	140	Traffic Sensor
T014	220	Traffic Sensor
T015	90	Traffic Sensor
T016	180	Traffic Sensor
T017	120	Traffic Sensor
T018	110	Traffic Sensor
T019	160	Traffic Sensor
T020	130	Traffic Sensor

Figure 28: Traffic Data by IoT Device Table

```sql
SELECT TD.Traffic_ID, TD.Vehicle_Count, ID.Device_Type
FROM Traffic_Data TD
JOIN IoT_Devices ID
ON TD.Device_ID = ID.Device_ID;

```

7.14 Location with High Vehicle Count

Location_Name	Total_Vehicles
Gulshan	120
Banani	200
Old Dhaka	150
Mirpur	110
Baridhara	180
Paltan	130
Farmgate	140
Shyamoli	220
Jatrabari	180
Khilgaon	120
Rampura	110
Savar	160
Aminbazar	130

Figure 29: Location and high vehicle count table

```
SELECT L.Location_Name, SUM(TD.Vehicle_Count) AS Total_Vehicles
FROM Traffic_Data TD
JOIN Location L
ON TD.Location_ID = L.Location_ID
GROUP BY L.Location_Name
HAVING Total_Vehicles > 100;

```

7.15 Locations where No Citizens Live

Location_Name
Paltan
Farmgate
Keraniganj
Jatrabari
Khilgaon
Rampura
Savar
Aminbazar
Kamalapur
Green Road
Malibagh
Lalbagh
Hatirjheel
Turag
Banasree
Agargaon
Ashulia
Motijheel
Old Dhaka
Baridhara
Mohakhali

Figure 30: Locations where no citizens live Table

```
SELECT L.Location_Name
FROM Location L
LEFT JOIN Citizen_Location CL
ON L.Location_ID = CL.Location_ID
WHERE CL.Citizen_ID IS NULL;

```

7.16 Counting Citizens in Each Area

Location Name	Total Citizens
Paltan	0
Gulshan	1
Bashundhara Residential Area	7
Tejgaon	1
Farmgate	0
Shyamoli	1
Keraniganj	0
Jatrabari	0
Khilgaon	0
Rampura	0
Savar	0
Banani	1
Aminbazar	0
Kamalapur	0
Green Road	0
Malibagh	0
Lalbagh	0
Hatirjheel	0
Khilkhet	1
Turag	0
Banasree	0
Agargaon	0
Dhanmondi	1
Ashulia	0
Motijheel	0
Old Dhaka	0
Uttara	1
Mirpur	1
Baridhara	0
Mohakhali	0

Figure 31: Counting citizens in each area and location table

```sql
SELECT L.Location_Name AS "Location Name", COUNT(CL.Citizen_ID) AS "Total Citizens"
FROM Location L
LEFT JOIN Citizen_Location CL
ON L.Location_ID = CL.Location_ID
GROUP BY L.Location_Name;

```

7.17 List of Citizens with Their Location and City

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	Location_Name	City
Md. Tanvir Hossain	Gulshan	Dhaka
Hasnat Karibul Islam Alindo	Bashundhara Residential Area	Dhaka
Md. Minhaujul Islam	Bashundhara Residential Area	Dhaka
Nur Ibne Kawsar Zitu	Bashundhara Residential Area	Dhaka
Sabbir Hossain	Bashundhara Residential Area	Dhaka
Ivy Khan	Bashundhara Residential Area	Dhaka
Meher Easha	Bashundhara Residential Area	Dhaka
Khaled Mahmud	Bashundhara Residential Area	Dhaka
Nasima Akhter	Tejgaon	Dhaka
Md. Abir Hasan	Shyamoli	Dhaka
Professor Dr. Kamruddin ...	Banani	Dhaka
Sayed Mahi Ashrafi	Khilkhet	Dhaka
Shahriar Kabir	Dhanmondi	Dhaka
Ayesha Rahman	Uttara	Dhaka
Taslima Begum	Mirpur	Dhaka

Figure 32: Citizens location and city table

```sql
SELECT CI.Name, L.Location_Name, L.City
FROM Citizen_Info CI
JOIN Citizen_Location CL
ON CI.Citizen_ID = CL.Citizen_ID
JOIN Location L
ON CL.Location_ID = L.Location_ID;

```

7.18 List Maintenance Activities by Department

Maintenance_Type	Status	Department_Name
Gulshan Link Road Repair	Completed	Transportation
Bank Building Maintainance	Ongoing	Finance and Budget
Bridge Inspection	Completed	Infrastructure
Hospital building repair	Planned	Finance and Budget

Figure 33: Maintenance Activities by Department table

```sql
SELECT IM.Maintenance_Type, IM.Status, D.Department_Name
FROM Infrastructure_Maintenance IM
JOIN Department D
ON IM.Department_ID = D.Department_ID;
```

```

```

7.19 Counting Maintenance Activities per Department

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

Department_Name	Total_Maintenance
Department of Head of the city	0
Defence and Public Safety	0
Infrastructure	1
Transportation	1
Environment and Sanitation	0
Urban Planning	0
Emergency Services	0
Health and Welfare	0
Education and Community Services	0
Finance and Budget	2
Technology and Innovation	0

Figure 34: Maintenance Activities per Department count table

```
SELECT D.Department_Name, COUNT(IM.Maintenance_ID) AS Total_Maintenance
FROM Department D
LEFT JOIN Infrastructure_Maintenance IM
ON D.Department_ID = IM.Department_ID
GROUP BY D.Department_Name;

```

7.20 List of Ongoing Maintenance in Each Location

Location_Name	Maintenance_Type	Status
Banani	Bank Building Maintainance	Ongoing

Figure 35: Ongoing maintenance in each location table

```
SELECT L.Location_Name, IM.Maintenance_Type, IM.Status
FROM Infrastructure_Maintenance IM
JOIN Location L
ON IM.Location_ID = L.Location_ID
WHERE IM.Status = 'Ongoing';
```

7.21 Location with Low Vehicle Count

Result Grid | Filter Rows: | Export: | Wrap Cell

Location_Name	Total_Vehicles
Dhanmondi	50
Motijheel	80
Uttara	90
Mohakhali	70
Tejgaon	60
Keraniganj	90

Figure 36: Location with low vehicle count table

```
SELECT L.Location_Name, SUM(TD.Vehicle_Count) AS Total_Vehicles
FROM Traffic_Data TD
JOIN Location L
ON TD.Location_ID = L.Location_ID
GROUP BY L.Location_Name
HAVING Total_Vehicles < 100;

```

7.22 Finding All Requests by Citizens Along With Their Names

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Request_ID	Citizen_Name	Request_Status	Request_Date
REQ001	Sabbir Hossain	Pending	2024-12-01
REQ002	Khaled Mahmud	In Progress	2024-12-02
REQ003	Taslima Begum	Completed	2024-12-03
REQ004	Ayesha Rahman	Rejected	2024-12-04
REQ005	Sabbir Hossain	Pending	2024-12-05
REQ006	Sayed Mahi Ashrafi	In Progress	2024-12-06
REQ007	Ivy Khan	Completed	2024-12-07
REQ008	Meher Easha	Pending	2024-12-08
REQ009	Md. Tanvir Hossain	Rejected	2024-12-09
REQ010	Nasima Akhter	Completed	2024-12-10

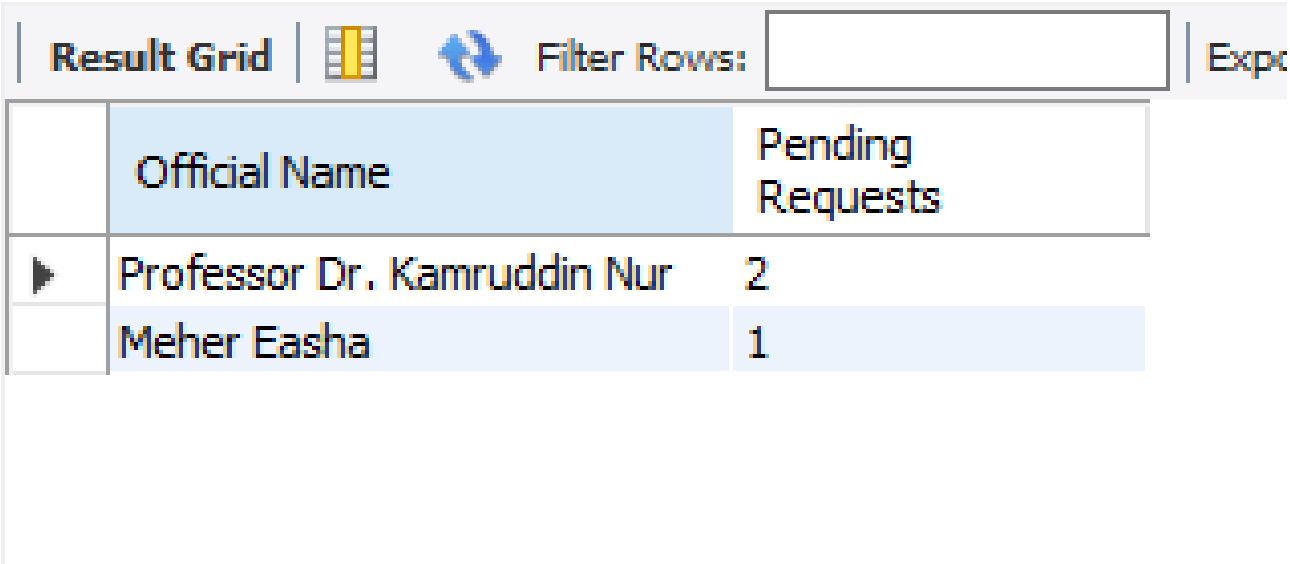
Figure 37: All Requests by Citizens Along With Their Names Table

```
SELECT SR.Request_ID, CI.Name AS Citizen_Name, SR.Request_Status, SR.Request_Date
FROM Service_Request SR
```

```
JOIN Citizen_Info CI
ON SR.Citizen_ID = CI.Citizen_ID;

```

7.23 Finding Officials with the Most Pending Requests



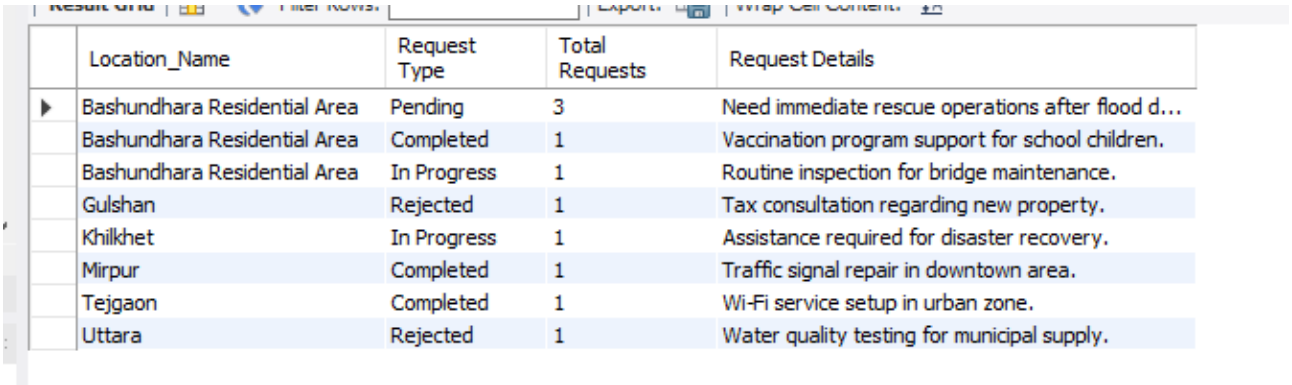
Official Name	Pending Requests
Professor Dr. Kamruddin Nur	2
Meher Easha	1

Figure 38: View all the Officials with the Most Pending Requests

```
SELECT CO.Name AS "Official Name", COUNT(SR.Request_ID) AS "Pending Requests"
FROM Service_Request SR
JOIN City_Official CO
ON SR.Assigned_Official_ID = CO.Official_ID
WHERE SR.Request_Status = 'Pending'
GROUP BY CO.Name
ORDER BY COUNT(SR.Request_ID) DESC;

```

7.24 Finding Locations with Highest Number of Requests



Location_Name	Request Type	Total Requests	Request Details
Bashundhara Residential Area	Pending	3	Need immediate rescue operations after flood d...
Bashundhara Residential Area	Completed	1	Vaccination program support for school children.
Bashundhara Residential Area	In Progress	1	Routine inspection for bridge maintenance.
Gulshan	Rejected	1	Tax consultation regarding new property.
Khilkhet	In Progress	1	Assistance required for disaster recovery.
Mirpur	Completed	1	Traffic signal repair in downtown area.
Tejgaon	Completed	1	Wi-Fi service setup in urban zone.
Uttara	Rejected	1	Water quality testing for municipal supply.

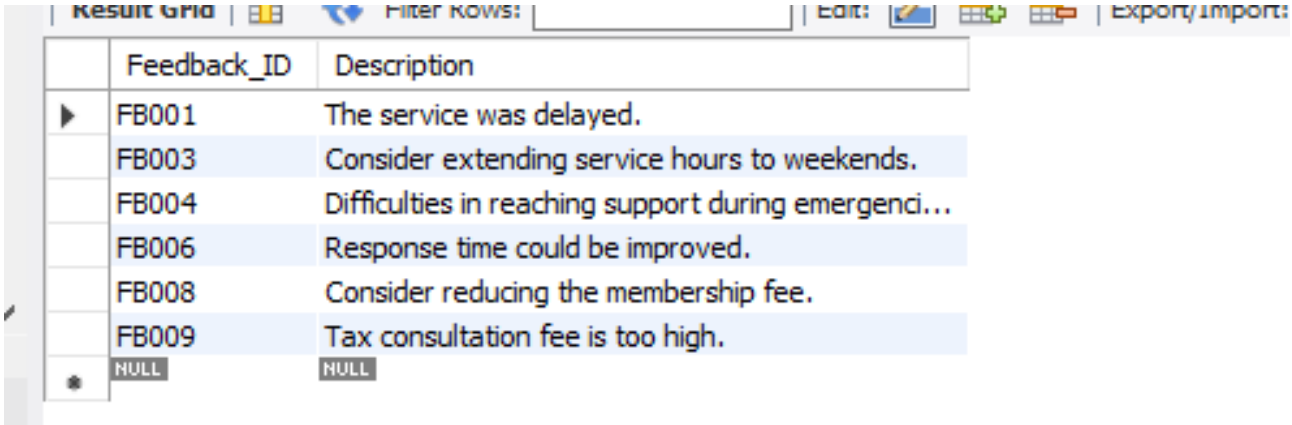
Figure 39: Locations with highest number of request Table

```
SELECT L.Location_Name AS "Location_Name",
       SR.Request_Status AS "Request Type",
```

```
       COUNT(SR.Request_ID) AS "Total Requests",
       GROUP_CONCAT(SR.Request_Description SEPARATOR '; ') AS "Request Details"
FROM Service_Request SR
JOIN Citizen_Location CL
ON SR.Citizen_ID = CL.Citizen_ID
JOIN Location L
ON CL.Location_ID = L.Location_ID
GROUP BY L.Location_Name, SR.Request_Status
ORDER BY COUNT(SR.Request_ID) DESC;

```

7.25 Retrieving Unresolved Feedback



Feedback_ID	Description
FB001	The service was delayed.
FB003	Consider extending service hours to weekends.
FB004	Difficulties in reaching support during emergenci...
FB006	Response time could be improved.
FB008	Consider reducing the membership fee.
FB009	Tax consultation fee is too high.
NULL	NULL

Figure 40: Unresolved Feedback Table

```
SELECT Feedback_ID, Description
FROM Feedback
WHERE Resolved_Status = FALSE;

```

7.26 List Available Services

Service Type	Service description
Emergency Rescue	Provide immediate rescue operations during nat...
Bridge Inspection	Conduct routine inspections and maintenance o...
Traffic Management	Monitor and regulate traffic flow across the city.
Urban Redevelopment	Plan and execute redevelopment projects in un...
Disaster Recovery	Coordinate recovery efforts after major disasters.
Vaccination Programs	Administer free vaccines for common diseases.
Library Membership	Provide access to public libraries and online reso...
Tax Consultation	Assist citizens in calculating and paying city taxes.
Wi-Fi Services	Offer free public Wi-Fi in selected urban zones.

Figure 41: Available Services Table

```sql
SELECT Service_Type AS "Service Type", Service_Description AS "Service description"
FROM Public_Services
WHERE Availability_Status = 'Available';

```

7.27 Services with a Fee Above 500 Taka

Service_Type	Service_Fee
Urban Redevelopment	10000.00
Tax Consultation	10000.00

Figure 42: Services with a Fee Above 500 Taka Table

```sql
SELECT Service_Type, Service_Fee
FROM Public_Services
WHERE Service_Fee > 500.00;

```

7.28 Services with a Fee Below 500 Taka

Result Grid | Filter Rows:

Service_Type	Service_Fee
Emergency Rescue	0.00
Bridge Inspection	0.00
Traffic Management	0.00
Water Quality Monitoring	200.00
Disaster Recovery	0.00
Vaccination Programs	0.00

Figure 43: Services with a Fee Below 500 Taka Table

```
SELECT Service_Type, Service_Fee
FROM Public_Services
WHERE Service_Fee < 500.00;

```

7.29 Counting Service Requests by Status

Result Grid | Filter Rows: | Export:

Service Request Status	Total Request Count
Pending	3
In Progress	2
Completed	3
Rejected	2

Figure 44: Counting Service Request by Status Table

```
SELECT Request_Status AS "Service Request Status", COUNT(*) AS "Total Request Count"
FROM Service_Request
```

```
GROUP BY Request_Status;

```

7.30 Finding Locations with the Most Frequent Services Requested and Assigned Official Details

Location	Total Request Count	Assigned Official	Role	Most Frequent Service
Bashundhara Residential Area	1	Professor Dr. Kamruddin Nur	City Mayor	Emergency Rescue
Mirpur	1	Professor Dr. Kamruddin Nur	City Mayor	Traffic Management
Uttara	1	Professor Dr. Kamruddin Nur	City Mayor	Water Quality Monitoring
Bashundhara Residential Area	1	Professor Dr. Kamruddin Nur	City Mayor	Urban Redevelopment
Khilkhet	1	Professor Dr. Kamruddin Nur	City Mayor	Disaster Recovery
Tejgaon	1	Professor Dr. Kamruddin Nur	City Mayor	Wi-Fi Services
Bashundhara Residential Area	1	Nur Ibne Kawsar Zitun	Officer	Bridge Inspection
Bashundhara Residential Area	1	Meher Easha	Officer	Library Membership
Gulshan	1	Ayesha Rahman	Senior Officer	Tax Consultation
Bashundhara Residential Area	1	Khaled Mahmud	Senior Officer	Vaccination Programs

Figure 45: Locations with the Most Frequent Services Requested and Assigned Official Details Table

```
SELECT L.Location_Name AS "Location",
       COUNT(SR.Request_ID) AS "Total Request Count",
       CO.Name AS "Assigned Official",
       CO.Role AS "Role",
       PS.Service_Type AS "Most Frequent Service"
FROM Service_Request SR
JOIN Citizen_Location CL
ON SR.Citizen_ID = CL.Citizen_ID
JOIN Location L
ON CL.Location_ID = L.Location_ID
JOIN City_Official CO
ON SR.Assigned_Official_ID = CO.Official_ID
JOIN Public_Services PS
ON SR.Service_ID = PS.Service_ID
GROUP BY L.Location_Name, CO.Name, CO.Role, PS.Service_Type
ORDER BY COUNT(SR.Request_ID) DESC;

```

7.31 List of Locations Experiencing Extreme Weather

Location	Weather Condition	Temperature	Wind Speed	Humidity
Motijheel	Stormy	18.70	45.00	85
Mirpur	Foggy	12.30	8.20	95
Shyamoli	Foggy	17.00	6.00	90
Old Dhaka	Foggy	15.50	5.00	90

Figure 46: Locations Experiencing Extreme Weather Table

```sql
SELECT L.Location_Name AS "Location",
       W.Weather_Condition AS "Weather Condition",
       W.Temperature AS "Temperature",
       W.Wind_Speed AS "Wind Speed",
       W.Humidity AS "Humidity"
FROM Weather_Condition_Data W
JOIN Location L
ON W.Location_ID = L.Location_ID
WHERE W.Wind_Speed > 50 OR W.Weather_Condition IN ('Foggy', 'Stormy')
ORDER BY W.Wind_Speed DESC, W.Humidity DESC;

```

7.32 Average Temperature and Humidity by Weather Condition

Weather_Condition	Avg_Temperature	Avg_Humidity
Windy	28.150000	65.0000
Sunny	27.980000	58.2000
Cloudy	25.780000	65.0000
Rainy	21.725000	80.0000
Stormy	18.700000	85.0000
Foggy	14.933333	91.6667

Figure 47: Average Temperature and Humidity by Weather Condition Table

```sql
SELECT Weather_Condition,
       AVG(Temperature) AS Avg_Temperature,
       AVG(Humidity) AS Avg_Humidity
FROM Weather_Condition_Data
GROUP BY Weather_Condition
ORDER BY Avg_Temperature DESC;

```

7.33 Monthly Average Weather Statistics

Result Grid | Filter Rows: | Export: | Wrap Cell Conten

Month	Average Temperature	Average Humidity	Average Wind Speed
12	23.775000	71.3000	14.620000

Figure 48: Monthly Average Weather Statistical Table

```sql
SELECT MONTH(Observation_Date) AS Month,
       AVG(Temperature) AS "Average Temperature",
       AVG(Humidity) AS "Average Humidity",
       AVG(Wind_Speed) AS "Average Wind Speed"
FROM Weather_Condition_Data
GROUP BY MONTH(Observation_Date)
ORDER BY Month;

```

7.34 City Official Details who Don't Have any Supervisors

Official ID	Name	Role	Deparment ID	Department Name	Contact Number	Email	Service years	Qualifications	Address
CF1	Professor Dr. Kamruddin Nur	City Mayor	1	Department of Head of the city	01777354323	kamruddin.nur@citygov.org	15	PhD in Ubiquitous Computing	House: 91; Road:04; Block:B, Banani Dhaka-1213

Figure 49: City Officials details who don't have any supervisors Table

```sql
SELECT CO.Official_ID AS "Official ID",
       CO.Name AS "Name",
       CO.Role AS "Role",
       CO.Department_ID AS "Department ID",
       D.Department_Name AS "Department Name",
       CO.Phone_Number AS "Contact Number",
       CO.Email,
       CO.Years_Of_Service AS "Service Years",
       CO.Qualifications,
       CO.Address
FROM City_Official CO
LEFT JOIN Department D
ON CO.Department_ID = D.Department_ID
WHERE CO.Supervisor_ID IS NULL;

```

7.35 Most Frequent Routes, Average Number of Stations and Status Distribution

Route	Total Buses	Average Stations	On Time Count	Delayed Count	Cancelled Count	Route Description
Route G	1	25.0000	1	0	0	Mirpur to Jatrabari
Route E	1	20.0000	1	0	0	Old Dhaka to Shyamoli
Route D	1	18.0000	0	0	1	Motijheel to Farmgate
Route J	1	16.0000	1	0	0	Paltan to Savar
Route A	1	15.0000	1	0	0	Gulshan to Paltan
Route H	1	14.0000	1	0	0	Baridhara to Khilgaon
Route B	1	12.0000	0	1	0	Banani to Bashundhara Residential Area
Route C	1	10.0000	1	0	0	Dhanmondi to Tejgaon
Route I	1	9.0000	0	0	1	Mohakhali to Rampura
Route F	1	8.0000	0	1	0	Uttara to Keraniganj

Figure 50: Most Frequent Routes Average Number of Stations and Status Distribution Table

```sql
SELECT BT.Route_Name AS Route,
       COUNT(BT.Bus_ID) AS "Total Buses",
```

```sql
        AVG(BT.Number_Of_Stations) AS "Average Stations",
        SUM(CASE WHEN BT.status = 'On Time' THEN 1 ELSE 0 END) AS "On Time Count",
        SUM(CASE WHEN BT.status = 'Delayed' THEN 1 ELSE 0 END) AS "Delayed Count",
        SUM(CASE WHEN BT.status = 'Cancelled' THEN 1 ELSE 0 END) AS "Cancelled Count",
        CONCAT(SL.Location_Name, ' to ', EL.Location_Name) AS "Route Description"
FROM Bus_Transportation BT
LEFT JOIN Location SL ON BT.Start_Location_ID = SL.Location_ID
LEFT JOIN Location EL ON BT.End_Location_ID = EL.Location_ID
GROUP BY BT.Route_Name, SL.Location_Name, EL.Location_Name
ORDER BY COUNT(BT.Bus_ID) DESC, AVG(BT.Number_Of_Stations) DESC;

```

7.36 Bus Route and Frequency Analysis

Route	Frequency Per Hour	Service_Level
Route G	10	High Frequency
Route E	8	High Frequency
Route D	7	High Frequency
Route H	6	High Frequency
Route A	6	High Frequency
Route J	5	High Frequency
Route B	5	High Frequency
Route I	4	High Frequency
Route C	4	High Frequency
Route F	3	Moderate Frequency

Figure 51: Bus route and frequency analysis Table

```sql
SELECT Route_Name AS "Route",
       Frequency_Per_Hour AS "Frequency Per Hour",
       CASE
           WHEN Frequency_Per_Hour >= 4 THEN 'High Frequency'
           WHEN Frequency_Per_Hour BETWEEN 2 AND 3 THEN 'Moderate Frequency'
           ELSE 'Low Frequency'
       END AS Service_Level
FROM Bus_Transportation
ORDER BY Frequency_Per_Hour DESC;

```

8 Conclusions

In the conclusion, Smart City Information Management System is the demonstration that shows the power of efficient data management and urban operations. The system facilitates smooth decision-making, increases transparency, and encourages citizen interaction by centralising a variety of city data, including transportation, utilities, and citizen services. With the scalable architecture and flexibility to include cutting-edge technologies, SCIMS is a solution that is prepared for the future and encourages efficiency and sustainability. SCIMS also lays the foundation for smarter and more connected cities.

9 Acknowledgments

We extend our sincere gratitude to our honorable faculty, **Prof. Dr. Kamruddin Nur**, for providing us with the opportunity to work on this project utilizing MySQL. Their guidance and support have been invaluable throughout development, enabling us to deepen our understanding of database management systems. We would also like to thank our project-mates for their collaborative efforts and dedication to this project. Together, we navigated through various challenges and milestones, leveraging our combined skills and knowledge to create a robust and functional database system.